A Comparative Analysis of Photonic Microcavity Architectures for Quantum-Dot Single Photon Sources: Performance, Trade-offs & Applications

Casey Beck, Iwan Cotgreave, Morgan McCabe, Morgan Parry-Williams, Jatin Patil, Antigonus Qin

Summary

A quantum-dot (QD) single-photon source (SPS) is useful only when the photon is delivered into the intended optical channel, mulitphoton emission suppressed and indistinguishability is preserved. Five nanophotonic architectures with a target resonance wavelength (λ_0) of 1300 nm are compared across common Figure of Merits (FoMs): micropillar, circular Bragg grating (CBG), Bragg Ring Micropillar (BRM), floating open cavity (FoC) and photonic crystal waveguide (PCW). Three-dimensional finite-difference time-domain (FDTD) modelled electromagnetic (EM) confinement, spectral response, and directional emission. A common source model then converted these into estimated phonon-weighted spectral composition, multiphoton probability and indistinguishability. None of architectures revealed universal superiority, though most-suitable-candidates were identified for the applications of Photonic Quantum Computing, Quantum Key Distribution, and Quantum Sensing & Metrology. Additional micropillar upper distributed Bragg reflector (DBR) pairs narrowed and strengthened the resonance, increasing Purcell enhancement and model-derived indistinguishability at the cost of finite-pulse re-excitation. CBG optimisation revealed two distinct operating regimes. Green Grass suppressed light escaping downwards to 0.15% and gave the lowest modelled $g^{(2)}(0) = 1.3 \times 10^{-3}$; and Blue Sky with Fp = 30.87 and 97.01% upward emission whilst being difficult to fabricate. Annular ring geometry in the BRM altered lateral leakage and broadened useful extraction by up to 65%, but what was suppressed laterally was not consistently directed upwards. The FoC combined strong simulated enhancement with post-fabrication tunability at greater alignement cost. The W1 PCW gave strong Purcell enhancement and guided-mode coupling, but the operating point near 994 nm meant guided outptut and convergence limits prevents a direct comparison to the other four architectures. An application-aware comparison revealed the BRM 2R geometry for practical photonic quantum computing (PQC), Green Grass for prepare-and-measure (P&M) quantum-key distribution (QKD) and photon-number sensitive metrology, and the FoC for Measurement-device-independent (MDI) QKD. Selection thus follows from intended application requirements and constraints with fabrication.

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Declaration

This project report is submitted towards an application for a degree at the University of Bristol. The report is based upon independent work by the candidates. All contributions from others have been acknowledged and the supervisor is identified on the front page. The views expressed within the report are those of the authors and not of the University of Bristol.

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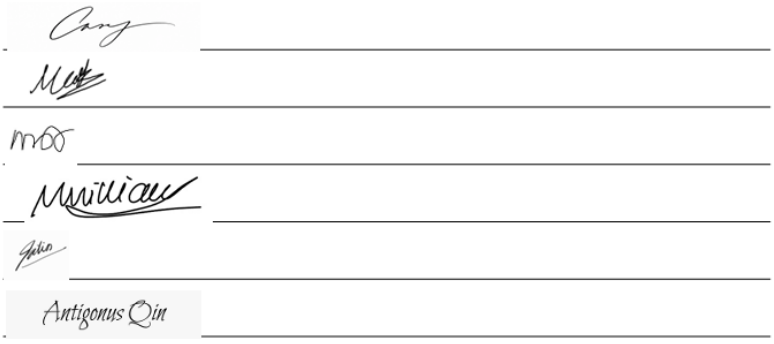

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Work Allocation

Contributor	Main contributions
CB:	Led BRM architecture & analysis. Created all analytical python scripts to calculate FoMs such as PSB contribution, purity etc Created comparison spider diagram. Contributed to project management by chairing meetings. Author of Chapter 7, co-author of Chapter 2, Chapter 9 & Chapter 10 & Chapter 11. Contributed to literature review, presentation and report-wide editing
MPW	Project Management Lead and Co-Group Leadership with CB; meeting organisation and work coordination/allocation; simulation and optimisation of the micropillar architecture; workflow and data validation for the hybrids; author of Chapter 5, co-author of the abstract, Chapter 1, 2, 10 and 11; literature review lead; report-wide and eLog book integration;
AQ:	Led the PCW architecture, simulation development, debugging, presentation and analysis. Author of Chapter 4 and co-author of Chapter 11.
MM:	Led the CBG architecture, simulation development, optimisation and analysis, including reference validation and development of the Green-Grass design. Initiated the experimental CBG design 'Blue-Sky'. Contributed to quantitative cross-architecture comparison. Author of Chapter 6 and co-author of Chapters 1, 2, 9 and 11.
IC	Worked with PCWs running simulation and initial reading/research, assisted in running simulation jobs. Also co-authoring Chapters 3,8 &11, and work on presentation introduction and conclusion sections.
JP	Led FoC architecture, analysis and presentation. Author of chapter 8. Co-author of Chapter 9

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1. Introduction - MPW & MM

1.1 Photonic Quantum Technologies and the Need for Single-Photon Sources

Photonic quantum technologies transmit, process and measure quantum information using controlled states of light. Light is an energy carrier that exchanges this energy in discrete amounts known as photons, and information can be encoded using its propagation path, polarisation or arrival time [1, 2]. Many photonic quantum operations require control of both the number of photons and the optical state they carry. An absent photon prevents the desired operation from occurring, whilst additional photons mean that the optical input is no longer a required single-photon state. Photons supplied from different times or sources must also be made sufficiently indistinguishable to interfere [1, 2].

To produce these controlled optical states, a single-photon source (SPS) is required to supply photons individually and repeatedly on demand. Ideally, each excitation generates one photon that is emitted directly into the useful optical mode. Figure 1.1 shows where the source sits within a wider photonic system. This project considers a source containing a semiconductor quantum dot (QD), which will be introduced further in Section 1.2.

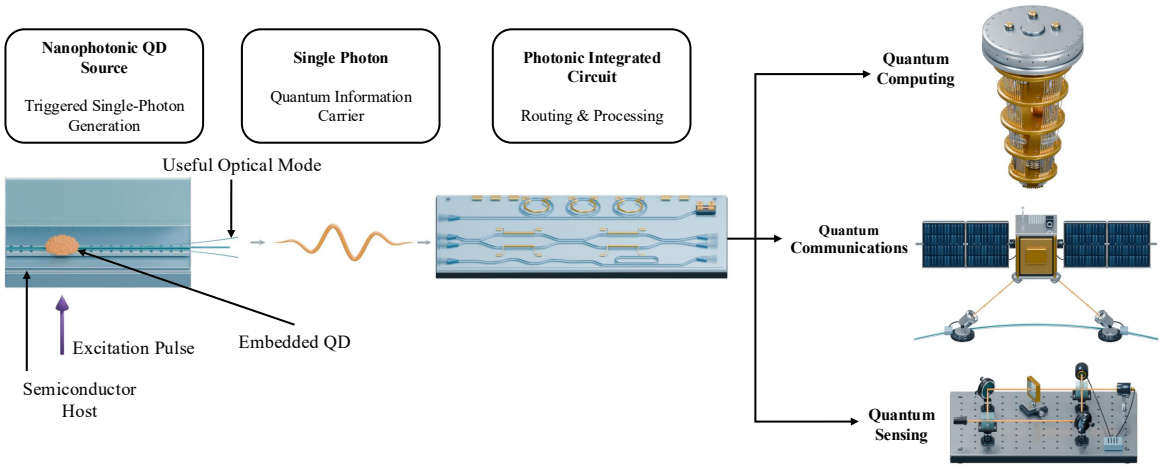


Figure 1.1. System-level role of a single-photon source in photonic quantum technologies. The source shown contains an embedded semiconductor quantum dot, introduced in Section 1.2. The "useful optical mode" denotes the optical channel received by the next component.

1.2 Semiconductor Quantum Dots and Single-Photon Emission

QDs are nanoscale regions of semiconductor formed within a larger semiconductor host. They possess discrete energy levels for electrons and holes (ie. negative and positive charge carriers) because their small dimensions confine carriers in all three spatial directions. In addition, the QDs considered in this project are epitaxially formed, with the difference in material composition between the dot and host producing band-edge energy barriers that further confine the carriers (an embedded region of lower bandgap material forms energy barriers which inhibit electrons and holes from moving away).

In all three spatial dimensions, carriers are confined to discrete energy levels rather than the quasi-continuous range available in bulk material [2, 3]. Photon emission can be initiated using optical excitation pulses (also known as triggers). These excite an electron to a higher-energy state within the QD, leaving behind a hole in the state they departed from. The electron-hole pair produced is confined entirely within the QD, allowing radiative recombination when their wavefunctions overlap. The photon emitted during this transition has energy corresponding to the gap between these states [2, 3], as illustrated in Figure 1.2.

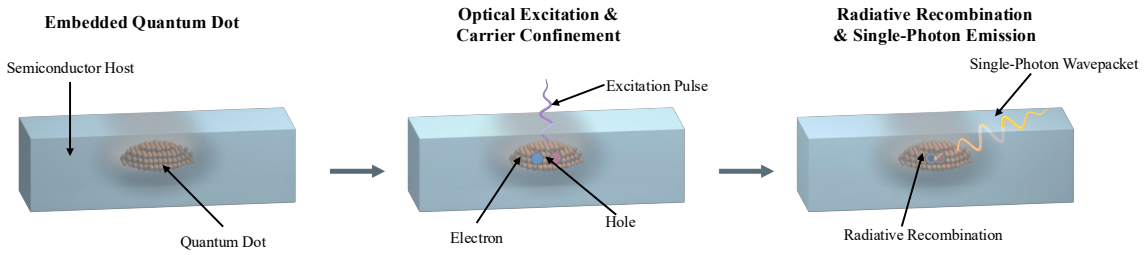


Figure 1.2. Schematic physical basis of triggered single-photon emission from an epitaxial semiconductor quantum dot. Optical excitation prepares a confined electron-hole pair within the embedded QD, radiative recombination releases that energy as an emitted photon. Following recombination, the transition must be excited again before another photon can be emitted. The schematic is not drawn to physical scale.

Recombination of the electron-hole pair empties this excited state, meaning it must be prepared again before the same optical transition can generate another photon. Once triggered, this behaviour provides the means to obtain one-photon-per-cycle operation. Realistic sources can deviate from this simple idealisation due to additional effects such as re-excitation, and therefore, the probability of multi-photon emission must be quantified. [2]. This is discussed in Chapter 2.

1.3 Requirement of a Useful Single-Photon Source

Photon generation alone does not define a useful SPS. The emitted photons must also be delivered efficiently into the required optical output, contain the intended photon number and be sufficiently indistinguishable between separate emission events [1, 2]. These properties describe different aspects of source performance and must therefore be considered seperatley

Efficiency describes the fraction of emitted light delivered into a stated useful output and is meaningful only when that output is clearly defined. Photon-number purity describes how effectively additional photons are suppressed within a single excitation cycle, while indistinguishability describes how similar separately emitted photons are. Differences in arrival time, optical frequency, polarisation or spatial mode can make photons distinguishable and reduce their ability to interfere [1, 2].

The relative importance of these properties depends on the intended application. Interference-based photonic systems place strong requirements on photon-number purity and indistinguishability, whereas other applications may place greater emphasis on efficient photon delivery. Spectral tunability is also important where the emitter must be aligned with a resonant optical structure. The investigated architectures are therefore evaluated according to their intended application rather than using a single measure of performance.

1.4 How Nanophotonic Structures Control Photon Emission

However a QD is embedded, its light will not spontaneously emit solely into the useful output. In an bulk semiconductor, QD emission can propagate in many directions, many of which away from the useful output. A semiconductor can be patterned & etched to make reflection, interference, scattering and optical guiding selective for particular output paths [4]. Such structures are referred to as Nanophotonic.

One common example is an optical Microcavity. Constructive interference between light reflected from cavity boundaries causes certain wavelengths to resonate, building up in intensity. The spontaneous emission rate from the QD is affected not only by its own properties but also by the optical fields present around the emitter at its transition frequency. One such field may be favoured at the QD position if a resonant cavity is present, allowing emission into that mode to occur at a faster rate. Enhancement in spontaneous emission rate due to modification of the local optical environment is known as the Purcell effect [4]. The cavity approximation and description of the local optical environment are presented in Chapter 2.

However, it should be noted that greater Purcell enhancement does not directly imply higher efficiency into the useful output. Photons can still be emitted in other directions and into unwanted optical modes. These differences are illustrated by micropillar research, in which high efficiencies into the useful output have been

credited to both resonant enhancement, and suppression of side and substrate emission [5, 6]. Nanophotonic engineering of the source must therefore control both how the QD emits and where the emitted light is delivered.

1.5 Nanophotonic Structures Investigated in This Project

Five nanophotonic structures are investigated, as shown below in Figure 1.3. They span waveguide-confinement, resonant confinement, radial scattering, combined cavity-grating control and tuneable open-cavity confinement, with established precedent in the QD literature [5–13]. Detailed theory is developed in Chapter 2.

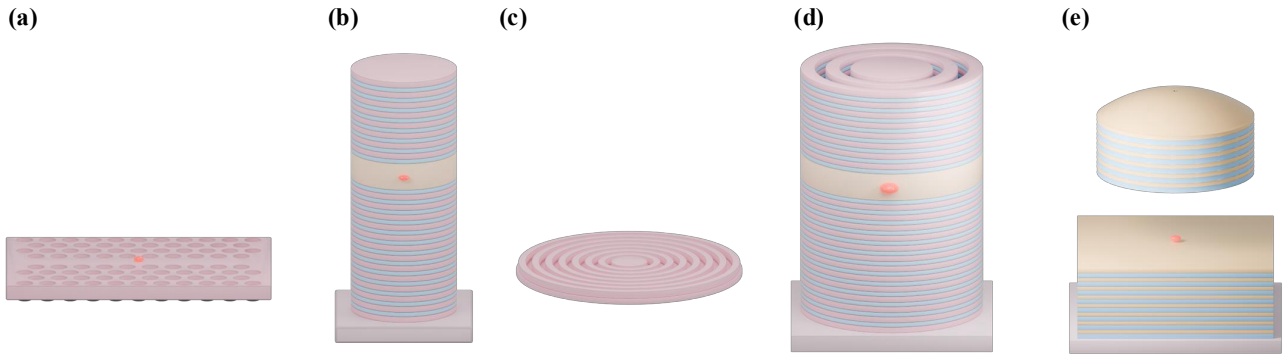


Figure 1.3. Nanophotonic structures investigated in this project: (a) W1 photonic-crystal waveguide, (b) micropillar, (c) circular Bragg grating, (d) hybrid micropillar-CBG, and (e) open cavity. The coloured marker indicates the approximate QD position. The structures are schematic and are not shown to a common physical scale.

A W1 photonic-crystal waveguide is formed from a thin semiconductor slab etched with a periodic array of holes. The periodic structure restricts in-plane propagation over selected wavelength ranges. Removing one row of holes creates a line defect along which light can propagate through the plane of the slab. QD emission can then be directed into a guided output rather than collected vertically [12, 13].

A micropillar forms a small vertical optical cavity around the QD. Distributed Bragg Reflector (DBR) stacks above and below the emitter are constructed from alternating semiconductor layers whose reflections reinforce over the required wavelength range. Repeated vertical reflection confines the optical field, while the finite pillar diameter also provides lateral confinement. Useful vertical output depends both on cavity enhancement and on suppressing light that escapes through side and substrate pathways [5, 6].

A circular Bragg grating (CBG) consists of a central emitter region surrounded by concentric rings. Light travelling radially from the QD is scattered by these rings, whose spacing is chosen so that the scattered contributions directed the field towards the desired direction [7–9].

The hybrid micropillar-CBG (later dubbed BRM) combines the preceding two mechanisms. A micropillar provides the central resonant cavity, while surrounding rings act on light that would otherwise leave laterally [10].

An open cavity confines light between two physically separated DBR stacks, with the topmost DBR stack having a concave geometry. Adjustment of the top DBR stack shifts the cavity resonance, allowing it to be aligned with the QD emission wavelength [11].

Although all five structures are intended to improve useful photon delivery, this is achieved through different physical mechanisms and output geometries and as such, performance must be considered with respect to each structure's intended function & use case.

1.6 Research Question, Aim, Scope & Objectives

The earlier sections of this chapter establish that useful SPS performance is multi-parameter and that the five structures investigated within this report mediate QD emission through different physical mechanisms. Therefore, a larger value of a single FoM cannot be equated to superior performance overall. In view of this, the central research question undertaken is:

> ***Which of the five nanophotonic structures provides the most useful SPS performance for different application requirements, and which physical mechanisms are responsible for those differences?***

The aim of this investigation is to answer this question through a controlled comparison of the micropillar, CBG, hybrid micropillar-CBG, open-cavity and W1 photonic-crystal-waveguide. Each structure is assessed using defined FoMs. A simulation-led approach is employed so that the architectures can be assessed under controlled emitter and numerical conditions. The propagation and confinement of the optical field are first calculated using the finite-difference time-domain (FDTD) method (Section 3.1), a full-wave numerical solution of Maxwell's equations, from which field distributions, spectral responses, radiated power and useful optical outputs are obtained directly. The objectives are to:

1. Develop and verify electromagnetic models of the five investigated Nanophotonic architectures.
2. Identify the physical mechanisms governing optical confinement, unwanted loss and useful photon delivery in each structure.
3. Improve each architecture within its investigated, numerically supported design space, using performance measures appropriate to its operating mechanism.
4. Use verified electromagnetic results to calculate defined FoMs.
5. Compare the resulting FoMs against stated application requirements.

Chapter 2 develops these concepts in greater detail and introduces the cavity and Nanophotonic theory required for the subsequent analysis. while Chapter 3 defines the EM modelling and verification methodology. Chapters 4-8 examine the individual architectures before the quantitative and qualitative comparisons are used to form the final project conclusions.

2. Theory & State of the Art MM & MPW

2.1 Single-Photon Purity and Coherence

Chapter 1 established that a useful source must suppress additional photons and produce photons that are sufficiently alike from one excitation cycle to the next. For the optical transition considered here, the QD is represented by an effective two-level system containing the ground state $|g\rangle$ and the exciton state $|X\rangle$. Real QDs also support charged and multiexcitonic states, in which more than one electron-hole pair can be present, but these are not required by the source model used here [2, 3]. Photon-number purity describes whether a nominal single-photon excitation cycle can also produce additional photons. It is quantified by the normalised second-order correlation function at zero time delay [14],

$$g^{(2)}(0) = \frac{\left\langle \hat{a}^\dagger \hat{a}^\dagger \hat{a}\hat{a} \right\rangle}{\left\langle \hat{a}^\dagger \hat{a} \right\rangle^2}, \tag{2.1}$$

where $\hat{a}$ and $\hat{a}^\dagger$ are the quantum-optical operators representing removal and addition of a photon in the detected optical mode. Their detailed operator algebra is not required for the later analysis; physically, $g^{(2)}(0)$ measures the tendency for multiphoton emission. An ideal single-photon source approaches $g^{(2)}(0) = 0$, while a non-zero value indicates residual multiphoton emission. The numerical procedure used to estimate this quantity under pulsed excitation is defined in Chapter 3. Indistinguishability depends on a different property which is optical coherence. Coherence describes how consistently emitted photons retain the same phase and frequency characteristics. After the exciton state is prepared, its radiative lifetime T_1 gives the characteristic timescale over which it decays and emits a photon. Pure dephasing can disturb the optical phase without requiring population decay, while slower spectral diffusion shifts the QD transition frequency between emission events [15, 16]. Either process can make successive photons less alike and reduce their ability to interfere, even when multiphoton emission is strongly suppressed. Consequently, a source can exhibit high single-photon purity while while still having limited photon indistinguishability. Leading to Photon-number purity and optical coherence must being treated as distinct properties. The finite-pulse, dephasing and phonon mechanisms underlying this distinction are considered in Section 2.4.

2.2 Purcell Enhancement and Cavity Response

Chapter 1 introduced the Purcell effect as a modification of spontaneous emission caused by the optical environment surrounding the QD. The QD can emit only into optical states available at its position and transition wavelength. A Nanophotonic structure changes this local environment and can favour some states while suppressing or redistributing emission into others. The availability of states around the emitter is described by the local density of optical states (LDOS) [4]. The resulting change in radiative emission rate is quantified by the Purcell factor. If Γ_{cavity} is the radiative decay rate in the Nanophotonic cavity and Γ_{bulk} is that in a bulk semiconductor (bulk meaning, a homogeneous 'block' of semiconductor material with no interfaces that would cause confinement or redirection of emission). then the Purcell factor is given by [4]

$$F_{\text{P}} = \frac{\Gamma_{\text{cavity}}}{\Gamma_{\text{bulk}}}. \tag{2.2}$$

Thus, $F_{\text{P}} > 1$ means faster radiative emission than in the bulk semiconductor, while $F_{\text{P}} < 1$ indicates inhibited spontaneous emission relative to bulk. When non-radiative processes are negligible, the radiative rate is inversely proportional to lifetime, so the same modification can be written as

$$F_{\text{P}} \simeq \frac{\tau_{\text{bulk}}}{\tau_{\text{cavity}}}. \tag{2.3}$$

For a cavity resonance, the available enhancement depends on how strongly the optical field is confined spectrally and spatially. Spectral confinement is described by the quality factor Q. As shown in Figure 2.1(b),

a resonance has a central wavelength λ_0 and a full width at half maximum (FWHM) $\Delta\lambda_{FWHM}$. For a narrow resonance,

$$Q \simeq \frac{\lambda_0}{\Delta\lambda_{FWHM}}. \tag{2.4}$$

A larger Q corresponds to a narrower resonance. This can strengthen the optical response close to λ_0, but it also narrows the wavelength range over which the QD remains well matched to the cavity. Spatial confinement is described by the effective mode volume V_{eff}. The cavity mode volume describes the effective space occupied by the optical mode and is calculated by integrating its energy density over space and dividing by its maximum value. A smaller V_{eff} therefore represents stronger concentration of the electromagnetic field.
Under ideal single-mode conditions, spectral and spatial confinement combine to determine the peak Purcell enhancement [4] this is given by (2.5):

$$F_{\text{P,max}} = \frac{3}{4\pi^2}\left(\frac{\lambda_0}{n}\right)^3 \frac{Q}{V_{\text{eff}}}, \tag{2.5}$$

where n is the refractive index of the medium housing the QD. (2.5) assumes weak emitter-cavity coupling, resonance between emitter and cavity, an emitter linewidth narrow relative to the cavity response, positioning at the field maximum and alignment of the transition dipole with the local cavity field. Weak coupling here means that emitted energy leaves through the optical mode rather than being coherently exchanged back and forth between emitter and cavity.
Real devices can depart from these conditions through spectral detuning, spatial displacement and dipole misalignment. Let the wavelength detuning between the QD and cavity be

$$\delta\lambda = \lambda_{\text{QD}} - \lambda_0. \tag{2.6}$$

If $\Delta F_{\text{P,cav}}(0)$ is the on-resonance Purcell contribution from the isolated cavity mode, its wavelength dependence close to resonance may be approximated by

$$\Delta F_{\text{P,cav}}(\delta\lambda) \simeq \frac{\Delta F_{\text{P,cav}}(0)}{1 + 4\left(\delta\lambda/\Delta\lambda\right)^2}. \tag{2.7}$$

Spatial displacement from the field maximum or poor dipole alignment further reduces the achievable resonant contribution [4]. Figure 2.1(c) illustrates the distinction between this resonant term and the total electromagnetic response. The isolated resonant contribution falls towards zero with increasing detuning, but background and leaky optical states remain available and may themselves be enhanced or suppressed. The total modification of emission therefore need not vanish away from resonance. This is particularly important for broadband micropillars, where useful output can arise from suppression and redistribution of competing decay channels as well as resonant enhancement [5, 6]. The full response extracted from FDTD includes these resonant and non-resonant channels together; its numerical definition and reference environment are specified in Chapter 3. Purcell enhancement describes how the radiative rate changes relative to that reference, but does not by itself specify where the emitted photon travels.

2.3 Architecture-Dependent Optical Behaviour

As introduced in Section 1.5, the five architectures control QD emission through different physical mechanisms. Consequently, their electromagnetic performance cannot be represented by a single common output quantity. The W1 photonic-crystal waveguide is primarily characterised by coupling into a guided mode [12, 13], whereas the CBG is accurately assessed through upward and finite-NA free-space extraction [7–9]. The micropillar and open-cavity structures are described more naturally through resonant confinement and the partition of the emission between useful output and undesirable radiation [5, 6, 11]. The hybrid structure is compared with its corresponding bare micropillar so that the additional contribution of the concentric grating can be isolated. These architecture-specific quantities and their numerical definitions are formalized in Chapter 3.

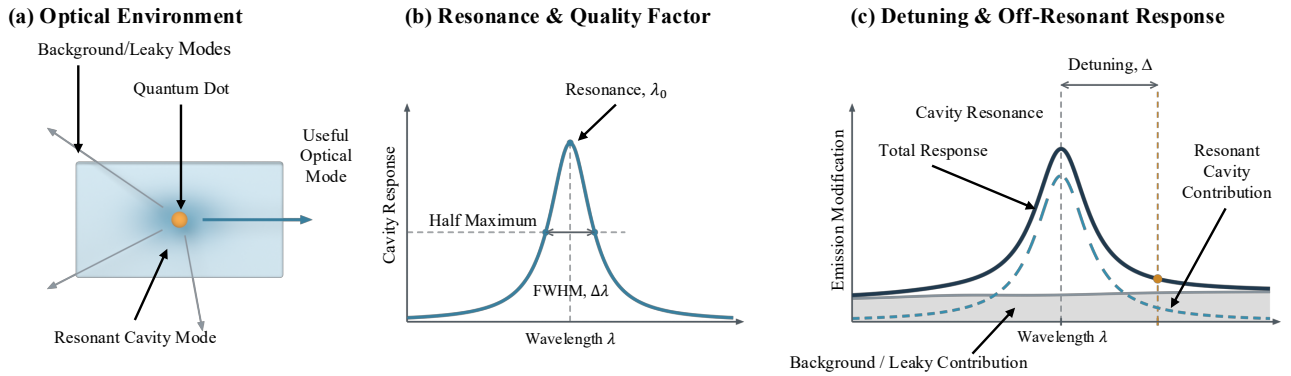


Figure 2.1. Cavity control of quantum-dot emission and resonant response. (a) Generic, architecture-independent schematic of a nanophotonic environment in which QD emission is redistributed between a resonant mode, useful output and background or leaky optical states. (b) The resonance wavelength λ_0 and FWHM $\Delta\lambda$ determine the quality factor $Q \simeq \lambda_0/\Delta\lambda$. (c) Wavelength detuning $\delta\lambda$ reduces the isolated resonant cavity contribution, while background and leaky states can maintain a non-zero total response away from resonance. Curves and pathways are schematic and not to scale.

2.4 Finite-Pulse Excitation, Dephasing and Phonons

Photon-number purity is also affected by the finite duration of the excitation pulse. Under resonant pulsed excitation, the QD may emit and return to the ground state before the pulse has ended. If sufficient excitation remains, the QD can be excited again within the same pulse, creating the possibility of an additional photon. This process, known as *re-excitation*, is illustrated in Figure 2.2 [14, 17].

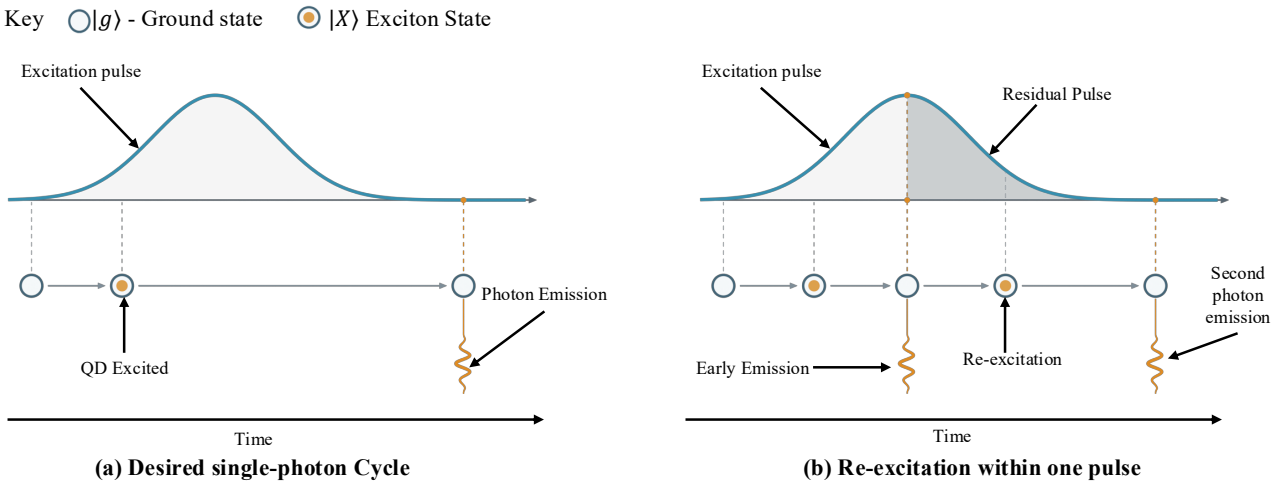


Figure 2.2. Finite-pulse excitation and re-excitation in a quantum-dot single-photon source. (a) Desired single-photon emission following pulsed excitation. (b) Re-excitation can occur if the QD emits while part of the excitation pulse remains, allowing an additional emission event within the same cycle. Timing is schematic and not to scale.

The radiative lifetime T_1 determines the timescale over which the excited QD emits. Purcell enhancement increases the radiative decay rate and therefore reduces T_1. Although faster emission is generally useful, for a fixed pulse duration it also makes it more likely that the QD emits before the excitation pulse has finished. The remaining part of the pulse can then excite the QD again. Therefore, the geometry with the largest F_P does not necessarily produce the lowest multiphoton probability. Pure dephasing and spectral diffusion instead affect the phase and frequency of the emitted photons [15, 16]. These effects can reduce the similarity between photons emitted in separate excitation cycles without producing additional photons within the same cycle. In the model used later in this report, finite-pulse re-excitation and phonon effects are included, while pure dephasing and slow spectral diffusion are not modelled separately. The calculated coherence-related quantities should therefore be treated as idealised. Phonons are quantised vibrations of the crystal lattice. Interaction between the confined carriers and longitudinal-acoustic phonons allows energy to be exchanged with the lattice [16, 18]. As a result, the emission spectrum contains a narrow zero-phonon line (ZPL), where

no net phonon energy is exchanged, and a broader phonon sideband (PSB) produced by phonon-assisted emission.
The nanophotonic structure can affect the ZPL and PSB by different amounts. If the ZPL is enhanced more strongly than the PSB, the fraction of useful emission contained within the ZPL increases. This is different from applying a passive filter after emission, since the structure changes the optical emission channels available to the QD itself. The temperature-dependent phonon model used in this work is described in Section 3.3.1.

2.5 State of the Art and Comparison Basis

The literature demonstrates high QD-source performance across several nanophotonic approaches, although reported values arise from different mechanisms and are defined at different optical boundaries. Three-dimensional broadband-micropillar modelling has predicted internal efficiencies above 90% together with indistinguishability above 99% under the definitions and assumptions of that study [6]. Here, *internal efficiency* refers to the photon-delivery boundary defined within that model rather than the overall photon-delivery efficiency of the complete system. Related micropillar work shows that such performance can result from suppression of background emission as well as resonant enhancement [5, 6], while resonantly driven micropillars have experimentally demonstrated high extraction efficiency together with near-unity indistinguishability [19].
CBG studies have demonstrated bright Purcell-enhanced emission and efficient free-space extraction, including at telecom wavelengths [7–9]. W1 photonic-crystal waveguides instead characterise photon delivery by coupling to the guided channel, with near-unity emitter-waveguide coupling demonstrated experimentally [12, 13]. Open microcavities offer adjustable spectral alignment between the QD and resonator [11].
These results show what each approach can achieve, but they cannot be compared directly using a single measure of performance. Internal efficiency, finite-angle collection efficiency and guided-mode coupling describe different stages of photon delivery and may use different starting powers, output boundaries and excitation conditions. Likewise, a large F_{P}, high Q, or small V_{eff} characterises the electromagnetic environment rather than the complete state or destination of the delivered photon.

Results obtained at different wavelengths must also be compared carefully, since efficiencies, linewidths and tolerances depend on the material system, refractive indices, emitter wavelength and device dimensions. Shorter-wavelength results can therefore guide the physical design but cannot be transferred numerically to the present telecom O-band structures without revalidation [7–9].

The remainder of this report therefore retains architecture-specific electromagnetic outputs while applying common definitions only where the underlying physical quantities are genuinely equivalent. Chapter 3 establishes these definitions and the common modelling procedure before the individual architectures are evaluated.

3. Methodology - CB & IC

3.1 Finite-Difference Time-Domain (FDTD) Method - CB, IC

Finite-Difference Time-Domain (FDTD) is a numerical analysis method used to model the propagation of electromagnetic fields through structures with spatially-varying material properties. This method has been used to simulate said propagation through each of the structures and as with any simulation/solving method has both merits as well as limitations which are discussed in section 3.1.4 The time-dependent Maxwell's equations in differential form, for a linear, isotropic, and source-free medium, are:

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t} \tag{3.1}$$

$$\nabla \times \mathbf{H} = \epsilon \frac{\partial \mathbf{E}}{\partial t} \tag{3.2}$$

Where $\mathbf{E}$ is the electric field vector (V/m), $\mathbf{H}$ is the magnetic field vector (A/m), ϵ is the permittivity of the medium, and μ is the permeability. In FDTD, these continuous spatial and temporal derivatives are substituted with discretised finite-difference approximations (3.3), converting the problem into a set of discrete equations that are able to be evaluated iteratively in time:

$$\frac{\partial f}{\partial x} = \frac{f\left(x + \frac{\Delta x}{2}\right) - f\left(x - \frac{\Delta x}{2}\right)}{\Delta x} + \mathcal{O}(\Delta x^2) \tag{3.3}$$

Wheer Δx is the spatial discretisation step and $\mathcal{O}(\Delta x^2)$ represents the discretisation error. FDTD simulations provide the primary numerical results used to assess the performance of the micro-cavities. The QD is digitally represented by an electric dipole source positioned within the cavity. The electromagnetic fields generated by the dipole source are modelled as waves propagating through various mediums.

3.1.1 Yee Cells

A Yee Cell (YC) is the fundamental structural unit of an FDTD simulation region (figure 3.1). They are unit cells and are arranged such that the boundaries of each cell align with the vector unit of the electric field($E_{x,y,z}$), an established convention[20].

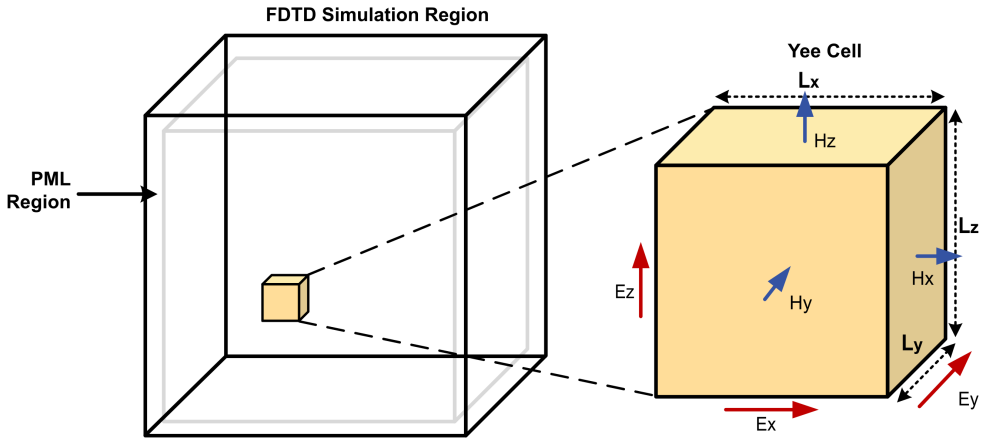


Figure 3.1. Illustration of Yee Cell within FDTD simulation region comprising of $E_{x,y,z}$ electric field and, $H_{x,y,z}$ magnetic field. Where L_x,L_y and L_z represent the side lengths of the cell and are not required to be equal.

The cells are comprised of a staggered mesh such that the magnetic field passes through the middle of the cell and the electric field is aligned as previously discussed. The YCs are simulation regions where Maxwell's equations equations (3.1) and (3.2) are solved by approximating the space and time derivatives as well as a leapfrog integration[20] to 'update' the fields. Ultimately, the YCs segment the region and allow the simulation to increment through the YCs to solve; it stands to reason, and is the case, that by increasing the number of cells, or as is commonly referred to as the mesh order, the higher the precision of the calculation. Specifically,

this is defined within the Lumerical program[1]. A higher Mesh order represents a higher density of cells and therefore, a smaller Δx (equation (3.3)) and subsequent higher precision, although this is a diminishing return.

3.1.2 Perfectly Matched Layers (PMLs)

Perfectly Matched Layers (PMLs) act as an artificial absorption region for propagating electromagnetic waves. With typical widths in the 10s of YCs, these layers exist on the outer boundary of an active FDTD simulation region as seen in figure 3.1. PMLs are markedly separate from a traditional absorbing material in that reflections at the interface are intentionally suppressed. As such, waves incident on a PML region are strongly absorbed, and can be effectively observed to be propagating infinitely into the material present at the FDTD region boundary.

3.1.3 Dipole Source

A QD, is the central source of emitted energy in the photonic microcavity, and is accurately represented as an oscillating electric dipole since its optical transition produces an electromagnetic field equivalent to that of a dipole emitter [21]. This is a useful representation as it allows for a classical analysis to be done accurately on what would ordinarily be a quantum system, that is the QD. This equivalence is know as the dipole approximation[22] An oscillating electric dipole (OED) is used by first placing the dipole in the optimum location for a QD, then due to the desire to simulate over a broad wavelength range a OED is more suited than the more narrow broadband of the QD when the main focus is with the cavity response. The relationship between the field intensity and single photon distribution can be used to predict efficiency [23].

3.1.4 Benefits and Limitations of FDTD

FDTD is a proven & suitable simulation platform, allowing for accurate electromagnetic field analysis of a photonic microcavity [24]. As is the case with all computational models, FDTD does have limitations that the results must be considered in context with (as discussed in section 4.5). A suitable mesh order (the density of Yee Cells within the simulation region) must be chosen to yield a sufficiently precise result, but not at the cost of computational resources. Simulation accuracy improvements decrease with each subsequent increase in Yee Cell density approximately exponentially, and as such, for lower order meshes a larger uncertainty is present.

3.2 Metric Extraction - IC

After performing FDTD calculations within Lumerical, the extracted data was refined to set of parameters that describe the behaviour of each structure such that they can be quantitatively assessed and compared. The direct data obtained from the simulation could be used to find a baseline efficiency from the transmissions out of the top, bottom and sides of the structure, as well as a measure of the Purcell factor with respect to wavelength. The transmission values, $\Gamma_{monitor}$ are defined in Equation 3.4.

$$\frac{\Gamma_{monitor}}{\Gamma} = \frac{P_{monitor}}{P_{source}} \tag{3.4}$$

Where Γ is the spontaneous emission rate of the QD in bulk Gallium Arsenide (GaAs), and can be expressed as $\Gamma = n\Gamma_0$ where $n \approx 3.4$ (the refractive index of GaAs at 1300 nm) and Γ_0 is the spontaneous emission rate of the QD in free space (itself able to be expressed as $\frac{1}{T_0}$ where T_0 is the radiative lifetime of a QD, which for the purposes of this project is selected as $T_0 = 1 \times 10^{-9}$s, as is a common standard.

Throughout this report, parameters such as Γ_{top}, Γ_{side} & Γ_{bottom} are used to quantify directional emission from the structures. These values, where presented, are expressed in terms of Γ (defined above). For example, a structure may report values in the range of (purely illustrative):

[1]Ansys Lumerical FDTD 2023 R1.3 is a commercially available FDTD modelling software. Simulations were conducted using BluePebble engine 8.29.3401 and external scripts for data extraction were written in python

$$\Gamma_{\text{top}} = 15\Gamma, \qquad \Gamma_{\text{side}} = 0.3\Gamma, \qquad \Gamma_{\text{bottom}} = 0.5\Gamma \tag{3.5}$$

3.2.1 Efficiency Factors; β, η, ξ

For each of the structures investigated in this project, a set of common metrics are adopted following nomenclature used by Dlaka [23] to model emitter-cavity coupling, top-directed outcoupling, and overall extraction efficiency. These are, in order, the active-coupling factor (β) (Eq. 3.6) describing the fraction of total emission coupling to the cavity-mode, the passive efficiency (η) (Eq. 3.7) which describes the fraction of the aforementioned cavity mode that is top-directed, and internal efficiency (ξ) (Eq. 3.8) which is the fraction of the total emission that is directed towards the top of the cavity (analogously, $\xi = \beta\eta$).

$$\beta = \frac{\Gamma_{\text{top}} + \Gamma_{\text{bottom}}}{\Gamma_{\text{total}}} \tag{3.6}$$

$$\eta = \frac{\Gamma_{\text{top}}}{\Gamma_{\text{top}} + \Gamma_{\text{bottom}}} \tag{3.7}$$

$$\xi = \beta\eta = \frac{\Gamma_{\text{top}}}{\Gamma_{\text{total}}} \tag{3.8}$$

where $\Gamma_{\text{total}} = \Gamma_{\text{top}} + \Gamma_{\text{bottom}} + \Gamma_{\text{side}}$, with each Γ representing the emission rate into the corresponding direction (Eq. 3.5).

3.3 Metric Processing - CB

3.3.1 Phonon Sideband Analysis

The PSB was modelled by implementing the effective phonon density of states (DOS) model $D_{PH}(\omega, T)$ used by Dlaka et al [6], which is based on the model presented by Madsen et al. [18]. The expression *(Eq. 3.9)* describes the frequency and temperature-dependent effective LA-phonon density of states experienced by a QD with defined material parameters, that is, the available LA phonon modes weighted by their deformation-potential coupling to the confined electron and hole states within the QD.

$$D_{PH}(\omega, T) = \frac{\hbar}{4\pi d c_l^5} \frac{\omega^3}{1 - e^{-\hbar\omega/(k_B T)}} \int_0^1 \left(D_e e^{\tilde{\omega}_{e,xy}^2(u^2-1) - \tilde{\omega}_{e,z}^2 u^2} - D_g e^{\tilde{\omega}_{g,xy}^2(u^2-1) - \tilde{\omega}_{g,z}^2 u^2} \right)^2 du \tag{3.9}$$

Where ω is the LA-phonon angular frequency, $\rho = 5370\text{kgm}^{-3}$ is the mass density of GaAs, and $c_l = 5110\text{ms}^{-1}$ is the longitudinal speed of sound in GaAs. The exponential terms ($e^{\sim}$) describe the effect of the finite spactial confinement of the QD electron wavefunctions as Gaussian form factors that suppress coupling to phonons in which the wavelengths become sufficiently small relative to the QD dimensions [18] Device temperature was set at $T = 4\text{K}$. The deformation potentials $D_e = -14.6\text{eV}$ and $D_g = 14.8\text{eV}$ describe, respectively, the coupling of the QD electron & hole states to lattice-strain. These values were chosen to remain consistent with the PSB model implemented by Dlaka Et al. [6].

The resulting phonon DOS determines the spectral shape of the PSB. Folowing Dlaka *et al.*, [6], the intrinsic QD spectrum was constructed from a narrow Zero-Phonon Line (ZPL) Lorentzian & the broadband Phonon Sideband (PSB) 2.4, with the bulk ZPL weight (defined as the fraction of the total emission spectrum that is contained within the ZPL) set to $W_0 = 0.9$ corresponding with Dlaka *et al* [6]. The intrinsic spectrum is then weighted by the structure's Purcell spectrum ($F_{\text{P}}(\omega)$) to attain the total emitted spectra ($S_{4\pi}(\omega)$). Additionally, the top emitted spectra ($S_{\text{top}}(\omega)$) is also calculated by weighting $S_{4\pi}(\omega)$ by the collection efficiency spectra ($\xi(\omega)$):

$$S_{4\pi}(\omega) = F_{\text{P}}(\omega) S_{\text{Bulk}}(\omega), \qquad S_{\text{top}}(\omega) = \xi(\omega) F_{\text{P}}(\omega) S_{\text{bulk}}(\omega), \tag{3.10}$$

from which, the top-emitted ZPL weight after cavity-weighting (W_{Top}) was calculated.

3.3.2 Indistinguishability I

Indistinguishability (I), expressed as a percentage, represents the degree of overlap in desired state between two photons. More simply, it measures the similarity of two separately emitted photons in all degrees of freedom (frequency/wavelength, polarisation, spatial mode, temporal mode). A strong I is particularly important

for certain applications that rely on two-photon interference (discussed further in Chapter 10). Following on directly from Section 3.3.1, assuming phonon scattering is the main source of distinguishability and that other mechanisms are comparatively negligible, I is able to be approximated as [6, 25]:

$$I \approx W_{top}^2 \tag{3.11}$$

3.3.3 Single-photon Purity $g^{(2)}(0)$ & $\mathscr{P}$

The single-photon purity for each structure was calculated using the short-pulse two-level system (2LS) approximation presented by Hanschke *et al.* [17]. For a resonantly-driven emitter, where the excitation pulse duration $\ll$ the radiative lifetime, the probability of re-excitation during the same pulse can be modelled as:

$$P_2 \approx \frac{\gamma T}{8} \tag{3.12}$$

Where P_2 is the probability of emitting two photons (subsequently, P_1 is the probability of emitting one photon). As such, an ideal SPS has $P_1 = 1$ and $P_2 = 0$. The corresponding single-photon purity is characterised by the zero-delay second-order correlation function (Chapter 2 Eq. 2.1) . Under single-photon operation, $P_1 \approx 1$, $P_2 \ll P_1$, and further higher-order emission probabilities are comparatively negligible. As such, the equation can be reduced to:

$$g^{(2)}(0) \approx \frac{\gamma T}{4} \tag{3.13}$$

Where $\gamma = F_{\mathrm{P}}\frac{1}{T}$, T being the QD radiative lifetime in a bulk semiconductor, assumed for these calculations to be 1ns (1×10^{-9}s), and where T is the shortest-allowed pulse-length supported by the cavity bandwidth. This value, in calculations, is obtained from the FWHM of the F_{P} spectrum. Finally, a purity value ($\mathscr{P}$) can be derived simply from:

$$\mathscr{P} = 1 - g^{(2)}(0) \tag{3.14}$$

which defines the degree to which multi-photon emission is suppressed as a percentage. This value is later used in Chapter 10 to quantify structure performance for selected applications.

3.3.4 Device Efficiency ϵ

Device efficiency (ϵ) is arguably a more rigorous method of quantifying the performance of a photonic microcavity. Extraction efficiency (ξ) (Chapter 2 Eq. 3.8) describes the fraction of optical emission directed through the desirable top channel at a given λ, however, given that QD emission is distributed across both the ZPL & the PSB, phonon-induced losses must be considered to provide a more accurate account of device performance. Following convention established by Dlaka *et al.* [23], a phonon-adjusted efficiency , namely ϵ, was calculated. The Purcell-weighted intrinsic QD spectrum ($S_{4\pi}(\omega) = S_{\mathrm{bulk}}(\omega)F_P(\omega)$) describes the spectrum emitted over all directions. Given that, the spectrum emitted into the useful top channel becomes

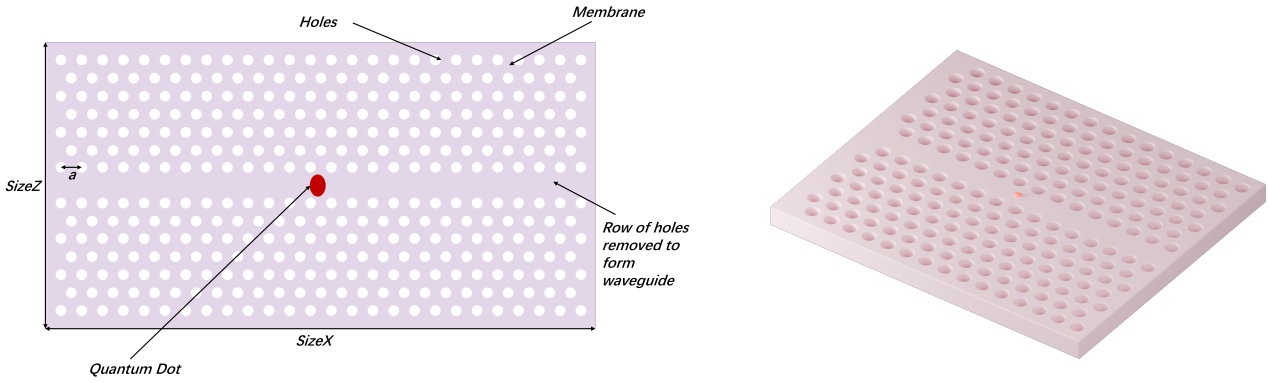
$$S_{top} = \xi(\omega)S_{4\pi}(\omega) \tag{3.15}$$

Where $\xi(\omega)$ is the top-efficiency (ξ) over a given range of frequencies (ω). Integrating both $S_{top}(\omega)$ and $S_{4\pi}(\omega)$ with respect to ω gives the spectral weight (the total relative emission across all frequencies) of the each spectrums. Device efficiency is therefore defined as the fraction of the total cavity emission that is directed through the top-channel:

$$\epsilon = \frac{\int S_{\mathrm{top}}(\omega)\, d(\omega)}{\int S_{4\pi}(\omega)\, d(\omega)} \tag{3.16}$$

4. Photonic Crystal Waveguides - AQ & IC

4.1 Architecture and Operating Principle - IC & AQ



(a) 2D schematic

(b) 3D model

Figure 4.1. A W1 photonic crystal waveguide, where a is the lattice constant and SizeZ/X are the lengths of the crystal used to define the number of rows or columns of holes within the photonic crystal. The gap shows the placement of the waveguide of length L.

Photonic crystal (PhC) structures can be broken into two main structure types; the first being cavity based structures, and the second being a waveguide structure 4.1. Both of these main ideas are broken down further into different types of each with much more variance in cavity based structures [26].
The Photonic Crystal Waveguide (PCW) was selected from the PhC family due to its strong optical confinement and its ability to couple emitted photons into a guided mode. This allows the structure to combine enhanced light–matter interaction with efficient photon extraction in the same device. The aim of this study was to evaluate whether a W1 PCW could provide strong Purcell enhancement and guided-mode coupling, and to identify the main numerical limitations of modelling this structure in 3D FDTD.
In its base form, a PCW is a photonic crystal generally made from a substrate that has holes arranged separated by a lattice constant a, with a quantum dot sitting among them and is generally grown, using such materials as GaAs, as is similar with other previously discussed structures. A waveguide 4.1 can be made by removing a line of holes, known as a light defect, and is where the light will couple to. Subsequently this can be transmitted down the waveguide for later use.
Despite the namesake PCW are devoid of a physical, separate waveguide, but rather a gap, or defect, within the crystal structure. This still operates like waveguide. In the case of PCWs the gap in-between the holes in the membrane functions as an effective higher refractive index region resulting in the entire structure acting as a waveguide with the coupled light propagating through it, as shown in figure 4.5. Conversely to something like a slab waveguide, a 2-D photonic band gap is created by removing or altering the holes to 'catch' the transverse magnetic and electric modes.
A PCW differs from cavity-based photonic crystal structures because its optical properties are mainly determined by the periodic geometry rather than by the material itself [26]. By changing this geometry, the propagation and extraction of photons can be controlled. PCWs can also provide a relatively broad operating range while maintaining strong Purcell enhancement compared with PHC cavities [27]. A PCW can operate near the photonic band edge, where the dispersion curve becomes flatter and the group velocity is reduced. This increases the local density of optical states (LDOS), ρ, as shown in Eq. 4.1, and can enhance the emission into the waveguide mode Γ_{wg}.

$$\rho \propto \frac{1}{v_g} \tag{4.1}$$

The increase of Γ_{wg} leads to a larger total coupling efficiency, β shown in 4.2 [26]

$$\beta_{wg} = \frac{\Gamma_{wg}}{\Gamma_{wg} + \Gamma_{loss} + \Gamma_{nr}} \tag{4.2}$$

With Γ_{loss}, and Γ_{nr} being the indicate the losses, such as non radiative as or emissions into other channels. In geometric terms, Γ_{loss} represents radiation that leaves through any non-waveguide channel, such as the top, bottom and side regions of the simulated structure. It should be noted that the β has an equivalence to ξ equation (3.8) that is demonstrated in 4.3.

$$\xi = \beta_{wg} \tag{4.3}$$

This is due to all light coupled into the waveguide is treated as extracted light, since the entire waveguide can be considered as the collection channel for the photons.
Whilst PCW have less variance than the cavity counterparts, there are still numerous types of PCW, commonly used and as such is the model this section is based from, is the W1 PCW. The W1 is defined to be a triangle lattice that has a single row of holes 'taken' out [26], [28] as demonstrated in figure 4.1 this is a type of linear defect created in 2D crystals. Another unique property of PCW is that they allow for chiral coupling [26],[29], meaning that circularly polarized photons can be directly coupled to the waveguide unlike it's counterparts.

4.2 Numerical Model Development and Debugging - AQ

Building the PCW model requires more repeated modification than the other architectures, as we had no mature PCW-specific FDTD template at the start of the project. Therefore, David Dlaka constructed the first version of the PCW model by adjusting the existing micropillar framework into the PCW geometry. The dipole source was placed at the centre of the line-defect waveguide for the simulations, except for the later QD position sweep.
However, the first set of simulations revealed two main problems: a large fraction of the transmitted power appeared to be side leakage, while the spectrum showed no clear resonance peak. It was hard to determine whether the problems originated from a single parameter or the structure itself, so further investigation with the literature and step-by-step debugging were needed.
The geometry reported by Patterson [30] was the first geometry that produced a clear resonance peak. However, the side leakage contribution was still unexpectedly large. Then we firstly checked whether the dipole orientation was causing poor coupling to the guided mode.

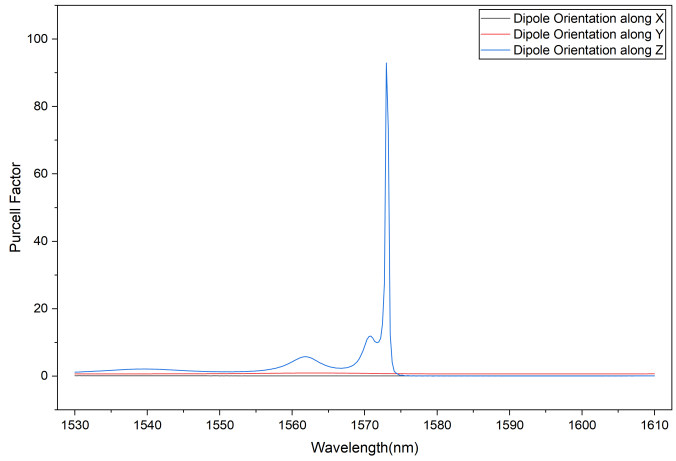
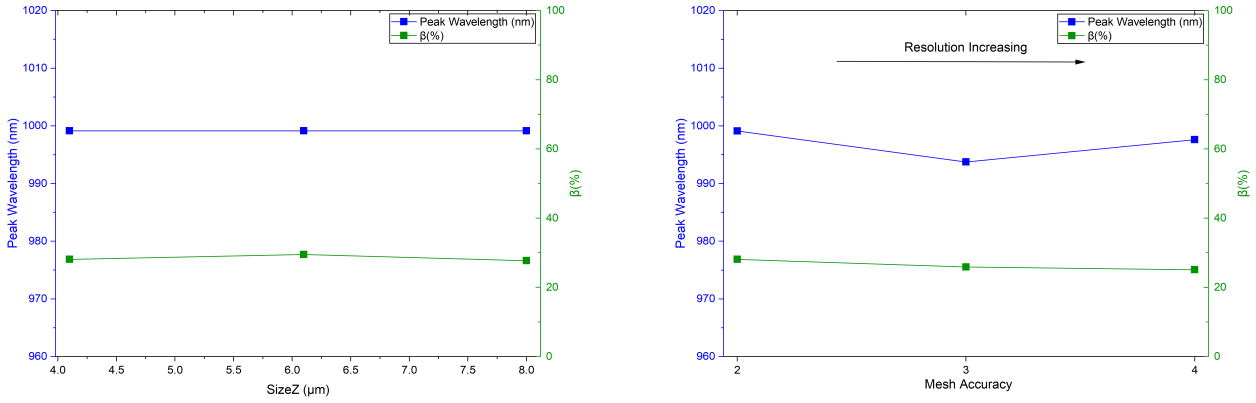


Figure 4.2. Purcell spectra for different dipole orientations.

Figure 4.2 shows that the Z-oriented dipole provides the best coupling result with clear resonance peak. This agrees with the polarization of the guided Bloch mode, whose electric field at the QD position is mainly aligned along Z. The other two dipole orientations couple only weakly to the mode and give almost no Purcell enhancement.
The Lund-Hansen W1 geometry [31] was then tested because the previous configuration still exhibited excessive side leakage and did not provide sufficient stability for further optimization. This provided a simpler reference structure for checking whether the side leakage problem came from the geometry or from the simulation setup. SizeX, SizeZ, mesh resolution and QD position were then swept separately.



(a) SizeZ comparison **(b)** Mesh Accuracy comparison

Figure 4.3. Comparison of the PCW results for (a) different transverse sizes, SizeZ, and (b) different mesh accuracy settings.

SizeZ (transverse size) was tried firstly, increasing from 4.1μm to 6.1μm and 8.0μm. As shown in Figure 4.3(a), the 6.1μm and 8.0μm cases produced almost identical spectra. SizeZ mainly controls how many photonic-crystal rows are included on each side of the line-defect waveguide, so increasing it effectively increases the transverse PhC cladding. Once enough rows are included to confine the mode, adding more rows has very little effect on the spectrum. The smallest converged value 4.1μm would be used in subsequent simulations to reduce computational cost.

Mesh accuracy was then increased from 2 to 4, where mesh accuracy controls the spatial discretisation of the FDTD simulation. As shown in Figure 4.3(b), the resonance wavelength shifted slightly, while the calculated β-factor remained almost unchanged. A coarser mesh slightly changed the shape of the circular hole, but this change was tiny relative to the hole radius. The effective r/a ratio remained nearly unchanged, so the outputs were similar. Considering both accuracy and simulation time, mesh accuracy 3 was used in the following simulations.

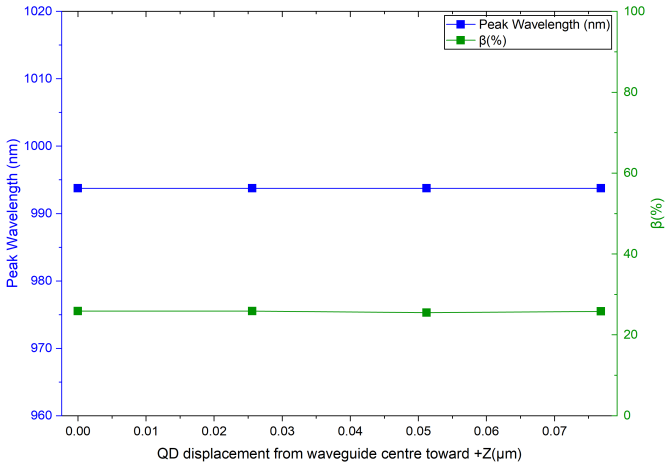


Figure 4.4. Comparison of the simulated response for different QD positions along the Z direction.

Then we checked QD position from the waveguide centre along the Z direction. As shown in Figure 4.4, varying the Z-coordinate from 0 to 0.3a produced limited changes in the simulated response. This suggests that the local guided-mode field varies only weakly over this displacement range, so the dipole–field overlap remains approximately unchanged. Lund-Hansen [31] also reported a relatively relaxed requirement on QD positioning in PCWs. But the result only showed weak sensitivity to Z-direction displacement within the tested range, rather than general insensitivity to QD position.

We then re-examined a previous field map to investigate the persistent side-leakage curve. As shown in Figure 4.5, the region between the white lines represented the central defect region, and part of the field

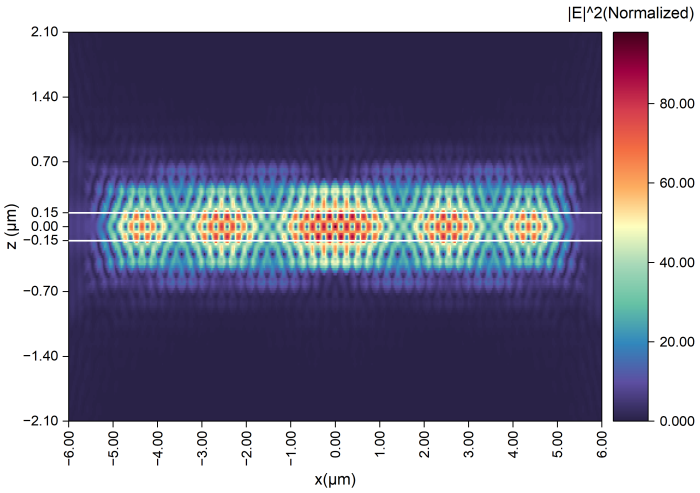


Figure 4.5. Electric-field distribution used to investigate the apparent side-leakage contribution.

propagated in the waveguide direction but through the surrounding photonic-crystal array rather than the central defect region. As it passed through the side monitors, the power of this field was counted as the side leakage even though it has the right direction.
The monitor width was increased to include the photonic-crystal region where the field propagates through. But to replicate the 100μm long paper-based structure would require an impractically long simulation time. A shorter SizeX sweep was then used as a practical substitute, with 8μm, 10μm, 12μm and 14μm long tested.

4.3 Electromagnetic Performance and Analysis - AQ

The 8μm and 10μm cases produced clear and usable resonances. In contrast, the longer structures exhibited slower field decay and higher-Q responses, preventing the simulation energy from reaching an acceptable auto-shutoff level within a practical runtime and resulting in jagged Lorentzian curve. In addition, the Purcell-factor also showed a large discrepancy from the total transmitted power. All of these phenomenon indicated insufficient convergence and made these longer-length results unreliable for quantitative analysis.
Here, the Q factor was used as a linewidth-derived quantity for the finite PCW resonance, calculated from $Q = \omega_0/\Delta\omega$, rather than as a conventional cavity-mode Q. The very large Purcell enhancement and narrow resonance may suggest that the system is approaching the strong-coupling regime. In this regime, the usual weak-coupling Purcell interpretation may no longer be valid. However, the present FDTD results are not enough to confirm this. Although the shorter structures could not fully reproduce the β-factor reported in the literature (89%), their converged responses were still suitable for performance evaluation. The 8μm and 10μm outputs could be compared to assess the Purcell enhancement, Q factor and β-factor.
Increasing SizeX from 8μm to 10μm shifted the main resonance from approximately 993.740 nm to 993.689 nm. As shown in Figure 4.6, the Purcell factor increased slightly from 764.1 to 771.8, while the β-factor decreased from 84.7% to 83.3%. The Q factor also showed increasing trend, rising from approximately 4.85×10^4 to 7.47×10^4.
The comparison between the 8μm and 10μm structures could not be evaluated as the dependence on SizeX. As X-direction is the propagation direction of the PCW, when SizeX was increased, it mainly increased the propagation distance rather than changing the local transverse geometry. The longer propagation distance may reduce the influence of the longitudinal boundaries and lead to a narrower resonance and higher Q factor. But the small changes in Purcell factor and β-factor are more likely related to the finite simulation length than to any change in the local PCW geometry.
As all light coupled into the waveguide is treated as extracted light, the overall collection efficiency is taken to be equivalent to the waveguide coupling efficiency, giving $\eta = 1$. The overall collection efficiency is then $\xi = \beta\eta$, which gives approximately 84.7% for the 8μm structure and 83.3% for the 10μm structure.

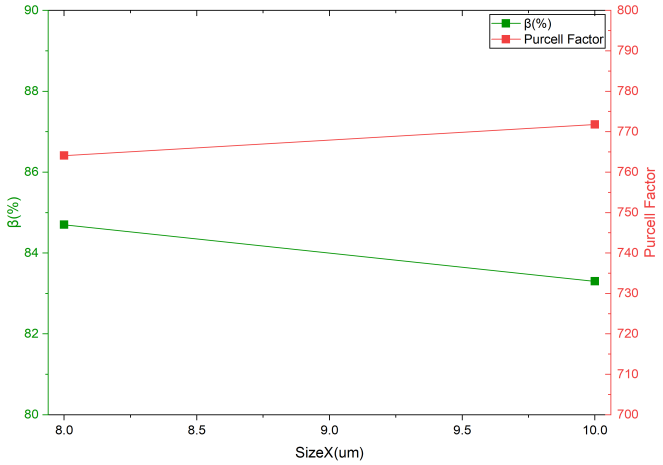


Figure 4.6. Comparison of the Purcell factor and β-factor for the 8μm and 10μm PCW models.

4.4 Advanced Analysis Discussion - AQ

The 8μm PCW structure was selected to assess whether the same advanced analysis about purity, the phonon sideband and the zero-phonon line can also be used on PCW.
Unlike the cavity-based architectures considered elsewhere in this project, the PCW does not rely on a single isolated cavity mode. Its Purcell enhancement is instead associated with the local density of optical states of the guided Bloch modes, while the extraction efficiency $\xi(\lambda)$ remains comparatively broadband and varies only weakly near resonance, as shown in Figure 4.7.

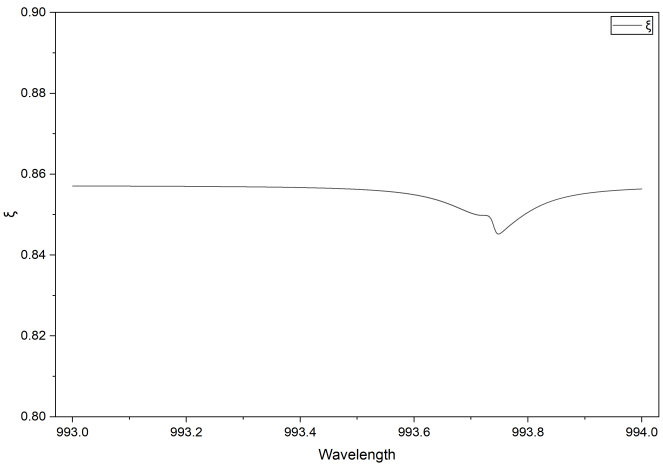


Figure 4.7. Wavelength-dependent extraction efficiency ξ of the 8μm PCW structure. The response remains comparatively broadband and does not exhibit a distinct extraction-efficiency peak at the narrow Purcell resonance.

As a result, the Hanschke-based purity workflow is difficult to apply directly as it requires a well-defined extraction-efficiency bandwidth to determine the effective pulse duration. The resulting purity estimate is not treated as a robust quantitative metric since no comparable ξ peak is present.
The broadband extraction response also means that off-resonant emission may still couple efficiently into the waveguide. This includes part of the phonon sideband as well as the zero-phonon line. While this can be favourable for total collection, coupling more phonon-assisted emission into the guided mode may reduce the fraction of indistinguishable photons. This means that the result depends not only on the total extraction efficiency, but also on the relative coupling of the ZPL and phonon sideband.
However, resolving this extremely narrow peak with sufficient data points while covering the wider spectral range would require a massive number of frequency points, further increasing the already high computational

cost of the simulation. In addition, using a fitted lorentzian curve to reduce the required spectral sampling could also suppress nearby weak spectral peaks, affecting the analysis.

4.5 Conclusions and Limitations - AQ

The PCW model produced a high-Purcell, high-β result, but the simulations also showed limitations in using 3D-FDTD for this structure. In particular, the extremely high Q factor led to very slow field decay. This meant that the simulation had to run longer before the residual energy reached the auto-shutoff level. When this condition was not reached, the Fourier-transformed spectra became jagged, and the Purcell factor data could not match the transmitted-power result. This situation restricted the number of converged simulations that could be completed within available time, especially for the longer waveguide structures. Lang also discusses this limitation of time-domain simulations, noting that the frequency resolution of FDTD is linked to the total simulation duration, whereas frequency-domain methods could determine individual eigenmodes [32].

A second limitation was the large number of parameters that affected the PCW response. Besides the QD position and dipole source orientation, the PCW response also depends on the lattice constant a, hole radius r, slab thickness h, selected row shifts and local hole displacement. Photonic-crystal designs are commonly described using normalised quantities such as r/a, h/a and a/λ [32]. When the material properties remain unchanged, Maxwell's equations are also scale invariant. To make a wavelength translation, the main geometric parameters need to be scaled together to guarantee that the dimensionless ratios are preserved. Although the geometric parameters should be scaled together in principle, unexpected results may still require individual parameters to be tested separately. Sweeping them independently can change several features of the photonic band structure at the same time. This also expands the parameter space and the scope of the optimisation.

The SizeX study was mainly used to test whether a shorter simulated waveguide could approach the behaviour of the much longer PCW. SizeX controls the longitudinal length of the PCW, so it was increased step by step while the geometry was kept unchanged. A longer SizeX would also increase the simulation time into an unacceptable level. It was not considered a conventional optimisation.

The peak is at 993.743 nm, outside the O-band used for the general comparison. The geometry was scaled towards the O-band while keeping the normalised ratios, but the scaled simulations did not produce a resonance peak. The next step would be to sweep the geometric parameters separately, making the parameter space much larger and requiring much more simulation time. In addition, the extremely narrow resonance and broadband extraction response made the common purity and phonon-related analysis difficult to apply reliably. For these reasons, the current PCW result was not used in the general comparison.

There is also a broader architectural distinction. The CBG, the micropillar, and the open-cavity are all rotationally symmetric structures based on vertical refractive-index modulation and DBR confinement. So the geometries, operating principles and principal optimisation variables share similarities for a common comparative framework. By contrast, the PCW uses an in-plane periodic array of air holes and relies on photonic-bandgap and slow-light effects for confinement. This also gives the PCW different spectral behaviour and makes some of the source-level metrics less directly comparable with the cavity-based structures. Photonic crystals themselves form a family containing standard W1 waveguides, modified W1 structures and other cavity-based PhC variants [26]. We would suggest that a future project treating photonic crystals as a separate design family and comparing several PhC architectures using the same wavelength, material platform and source-performance metrics.

5. Micropillar Optimisation - MPW

5.1 Architecture, Reference Basis and Optimisation Strategy

A micropillar consists of a cylindrical optical cavity fabricated around the emitter such that spontaneous emission funnels into one confined resonance rather than radiating freely in all directions. Distributed Bragg Reflectors (DBRs) at the top and bottom of the pillar confine the field vertically through their alternating layers of different refractive indices which act as wavelength-selective mirrors, while the pillar sidewall etched into a cylinder confines the light laterally. Taken together, these boundaries limit the number of optical modes accessible to the QD and preferentially enhance emission into that mode of the resonant cavity [33]. The confinement is not perfect however, and photons captured into competing radiation channels are lost from the useful signal, so increasing the resonant enhancement of the pillar does not automatically mean a larger useful optical output [5, 6, 34]

Two geometrical parameters were tuned, chosen because they act on different parts of the cavity response. Adding layers to the upper DBR increases the reflectivity of the top mirror thereby increasing the strength of the resonance and modifying the ease with which light can couple out of it. The pillar diameter D determines the lateral confinement; adjusting it changes the spatial shape of the mode along with its resonance wavelength, and ratio of emission into the desired axial radiation channel compared to all other channels. As neither effect linearly scaled with measured source performance, both were optimised using simulation rather than predetermined.

Four micropillar families were investigated, comprising 7, 9, 11 and 13 upper DBR pairs and denoted 7TP–13TP. Each retained the same 35.5-pair lower DBR, QD position and overall material system, while their upper reflectors and optimised diameters differed. Any given family is therefore not one structure with a single variable altered, but rather each represents a separately optimised design. Conventions used for the common emitter, monitor and numerical analysis in Chapter 3 were continued here.

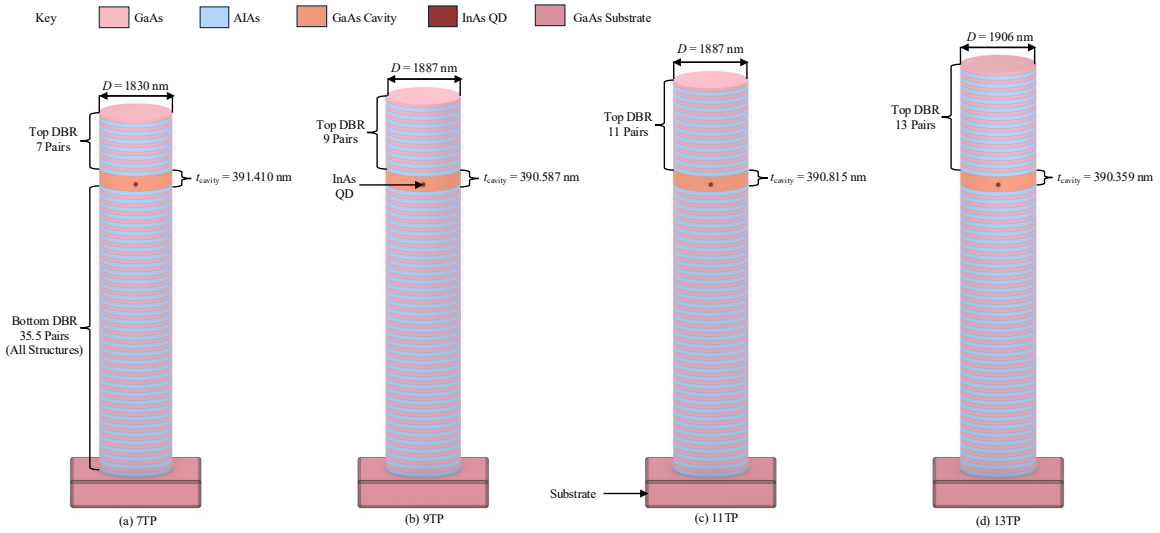


Figure 5.1. Final geometries of the 7TP, 9TP, 11TP and 13TP micropillar families. All structures retain the common 35.5-pair lower DBR and QD position. D denotes the pillar diameter and t_{cav} the cavity thickness; the labelled dimensions correspond to the geometries selected for each family.

Optimisation was performed in two stages. The diameter of the pillars was tuned first as this controls both lateral confinement as well as how the emitted power is distributed. Once an optimum diameter was found for each family of pillars this diameter was kept fixed and the cavity thickness t_{cav} was tuned instead so that the resonance could be moved towards the target wavelength of 1300 nm. Separating lateral optimisation from spectral tuning allows us to compare the final families of micropillars without forcing identical lateral geometries on physically different resonators.

5.2 Diameter Optimisation across 7/9/11/13TP Families

The pillar diameter was optimised independently for each DBR family. Moving the cylindrical sidewall affects multiple characteristics of the cavity response simultaneously. The confinement of the optical field in the lateral direction is changed, along with its resonance position and the ratio of emitted power through the intended vertical output versus other radiation channels. The diameter with maximum resonant enhancement therefore does not necessarily correspond to the diameter that gives the greatest upward emission [5, 6, 33, 34].

Three simulated quantities are used to analyse the diameter sweeps. F_{fit} is the fitted peak-enhancement quantity used in this step of the optimisation. Γ_{top} is the top-directional emission rate calculated through the upper monitor, using the monitor-based extraction method outlined in Section 3.2. The corresponding internal efficiency ξ, outlined in Section 3.2.1, is the fraction of the total monitored emission directed through the top channel. It can be further separated into that fraction coupled into the two axial channels β and the fraction of the axial emission that travels up vs down, η. These parameters therefore serve to distinguish the strength of cavity effect on emission vs where that emission ends up being directed.

Family	D at max. F_{fit} (μm)	D at max. Γ_{top} (μm)	D at max. ξ (μm)	Selected D^* (μm)
7TP	1.91	1.83	1.83	1.83
9TP	1.83	1.83	1.91	1.89
11TP	1.87	1.91	1.89	1.89
13TP	1.87	1.87	1.89	1.91

Table 5.1. Diameter-selection evidence for the four micropillar families. The first three columns identify the sampled diameter at which each decision metric reached its maximum; D^* is the diameter retained for subsequent cavity-thickness tuning.

The separation of these maxima explains why diameter could not be chosen based on a single figure-of-merit. For 7TP, Γ_{top} and ξ both peaked at 1.83 μm, despite the fitted cavity enhancement being largest at 1.91 μm. This highlights that strong cavity enhancement and high useful upward power are not comparable goals. For 9TP, there was no single sampled diameter which maximised all three quantities. Instead, 1.89 μm was kept as the engineering compromise between them across the sampled response. This diameter was also used for 11TP because it maximised ξ while retaining strong upward emission.

For the 13TP, the local sweep gave no sufficient reason to move away from the established 1.91 μm geometry, hence this retention rested on weighted evidence, not just a proven optimum. These accepted diameters range from only 1.83–1.91 μm, but their variation with device indicates that the preferred lateral boundary condition depends on the entire DBR stack. Each D was therefore fixed prior to the cavity-thickness optimisation, allowing free variation of spectral position without re-opening the lateral design.

5.3 Cavity-Thickness Optimisation and Spectral Alignment

Selecting the pillar diameter does not ensure that the cavity resonance remains at the 1300 nm design wavelength. This is important because the Purcell response is strongly wavelength dependent: if two structures were compared with substantially different detunings, some of the differences in resonant enhancement could be attributable simply to one cavity being better spectrally matched. The cavity thickness was therefore adjusted after diameter selection to bring each resonance back towards 1300 nm.

For each family, D^* was fixed while t_{cav} was varied and the resulting wavelength-dependent Purcell response was evaluated using the EM extraction framework of Section 3.2. Over the sampled thickness range, the fitted resonance wavelength changed approximately linearly with t_{cav}, providing a predictable local means of spectral tuning.

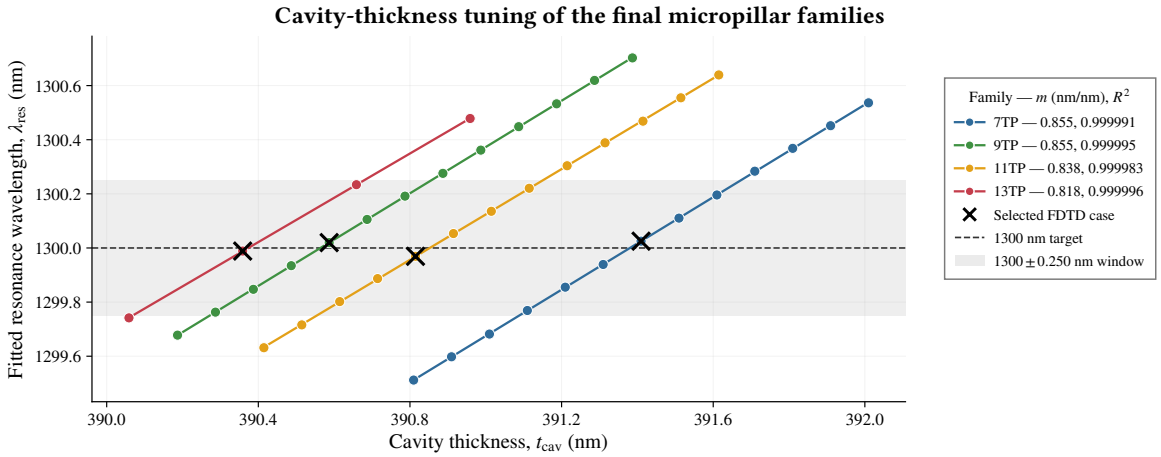


Figure 5.2. Cavity-thickness tuning of the 7TP, 9TP, 11TP and 13TP micropillar families during the optimisation stage. Each coloured series shows the fitted resonance wavelength obtained as t_{cav} is varied at fixed D^*. Black crosses identify the selected simulated cases, the dashed line marks the 1300 nm design target, and the shaded region indicates the ±0.25 nm acceptance window.

Family	Tuning gradient (nm/nm)	Selected t_{cav} (nm)	Fitted λ_{res} (nm)	Detuning from 1300 nm (nm)
7TP	0.85	391.41	1300.03	+0.03
9TP	0.86	390.59	1300.02	+0.02
11TP	0.84	390.81	1299.97	−0.03
13TP	0.82	390.36	1299.99	−0.01

Table 5.2. Cavity-thickness tuning and optimisation-stage spectral alignment. The tuning gradient gives the local change in fitted resonance wavelength per nanometre change in cavity thickness. All selected values correspond to explicitly simulated geometries.

All four families detune in the same direction: thickening the cavity shifts the resonance toward longer wavelength, at 0.82–0.86 nm, so a 1 nm change in cavity thickness corresponds to approximately 0.8–0.9 nm of resonance shift across the explored parameter space. The tight grouping between families indicates that the resonance wavelength is set by properties of the full optical cavity rather than only the central layer. Since both D^* and the upper DBR differ between each family, the trend cannot be explained by TP count alone.

Every accepted geometry lies within approximately 0.03 nm of the 1300 nm target at this stage. All of the points that have been retained are cases explicitly simulated during optimisation: the fitted functions were used to guide the search, but were never used to fill in a design instead of producing an FDTD result. The listed wavelengths at this stage of optimisation are therefore not the final values; each accepted design was re-simulated at much higher spectral resolution as part of a validation stage described in Section 5.4.

5.4 Final Electromagnetic Comparison

Accepted geometries were re-simulated at higher spectral resolution. This validation step does not re-open either optimisation and simply provides the final spectra and directional-emission quantities upon which the four designs will be compared. The most immediate result is that the final resonances occur close to 1295 nm, rather than at the 1300 nm positions obtained during the cavity-thickness optimisation of Section 5.3. The fitted resonance wavelengths are 1294.98, 1295.07, 1295.13 and 1295.02 nm for 7TP–13TP, respectively. The shift from the design target is therefore similar for every family, while the final resonances themselves span only 0.15 nm. This preserves close spectral alignment between the four designs for the purpose of comparing their resonant behaviour. The origin of the approximately 5 nm shift was not isolated in the available FDTD simulations, and thus is treated as a numerical observation rather than given a physical mechanism.

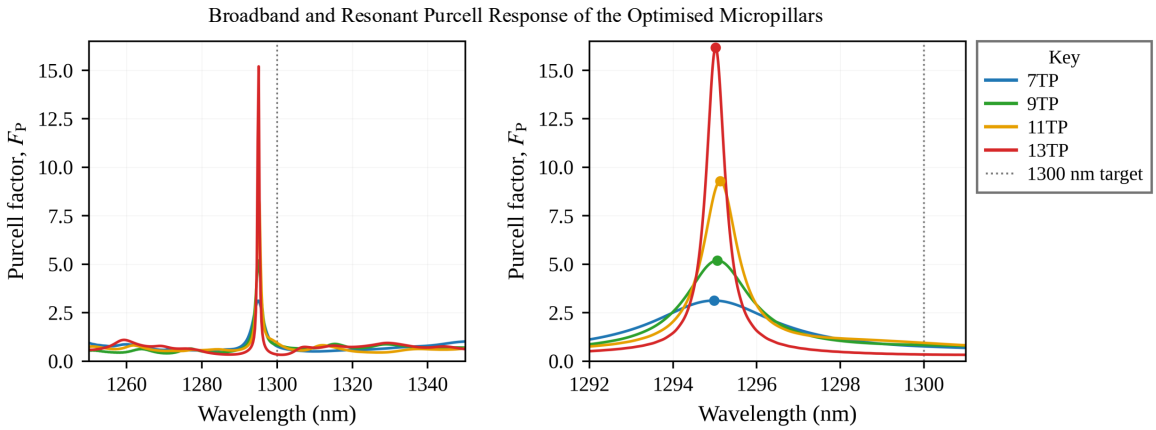


Figure 5.3. Purcell spectra of the four optimised micropillar designs. The left panel shows the broadband response, while the right panel resolves the cavity resonances using the densely sampled final spectra. Coloured markers indicate the directly sampled Purcell maxima; the values reported in Table 5.3 are obtained from fits to the corresponding resonance. The dotted vertical line marks the nominal 1300 nm design target, not the fitted resonance position. Final validation places all four resonances close to 1295 nm

The Purcell factor quantifies the extent to which the local optical environment alters the QD radiative rate, while Q is a measure of how spectrally narrow the cavity resonance is, as established in Section 2.2. Across the final designs, the peak Purcell factor rises from 3.13 for 7TP to 16.13 for 13TP. Over the same progression, Q increases from 365.81 to 2349.19, while the FWHM narrows from 3.54 to 0.55 nm. Therefore, the higher-TP designs exhibit substantially stronger and more spectrally selective resonant enhancement. This trend is consistent with enhanced axial confinement as the upper mirror becomes more reflective [33].

Family	λ_{res} (nm)	Peak F_P	Q	FWHM (nm)	β	η	ξ
7TP	1294.98	3.13	365.81	3.54	0.93	0.94	0.88
9TP	1295.07	5.20	694.12	1.87	0.98	0.92	0.90
11TP	1295.13	9.28	1287.18	1.01	0.98	0.94	0.92
13TP	1295.02	16.13	2349.19	0.55	0.99	0.97	0.96

Table 5.3. Final electromagnetic performance of the four optimised micropillar designs. λ_{res}, peak F_P, Q and FWHM are obtained from fits to the densely sampled final resonances. Q is calculated from the full-precision fitted resonance and linewidth before report-facing rounding. The directional quantities β, η and ξ, defined in Section 3.2.1, are evaluated at the fitted resonance wavelength.

A rising Q is not an unconditional improvement. Concentrating the same enhancement into a narrower band makes the response proportionally more sensitive to detuning, hence the strongest cavity is also the least tolerant, which is why the offset above weighs more heavily on 13TP than on 7TP. Table 5.3 accordingly characterises each structure at its own fitted resonance rather than at exactly 1300 nm.

The directional quantities improve on a very different scale. Recalling Section 3.2.1, β is the fraction of total emission entering the two axial channels, η the fraction of that travelling upwards, and the fraction $\xi = \beta\eta$ reaching the useful top channel. Across the four families, ξ rises only from 0.88 to 0.96, i.e a 9% relative gain set against a fivefold increase in peak enhancement: making the cavity resonate more strongly does not deliver a proportionate amount of extra light upwards. The dip in η at 9TP before its recovery at 13TP is retained as an observed feature; the available directional decomposition does not establish a unique mechanism for it.

Figure 5.4 provides a separate check that the same kind of mode is being compared throughout. All four field distributions remain centrally localised with a single lobe, consistent with the fundamental HE_{11}-like micropillar mode [33], so the rise in enhancement is not accompanied by a transition to a different transverse mode.

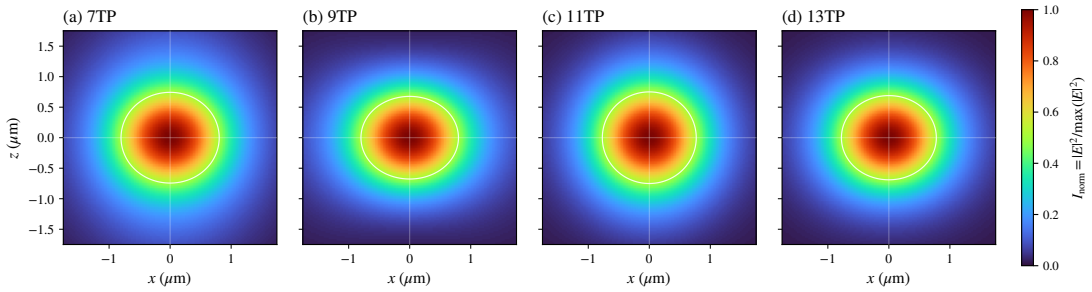


Figure 5.4. Normalised resonant longitudinal electric-field intensity distributions for the final 7TP, 9TP, 11TP and 13TP micropillar geometries. Each panel shows $I_{\mathrm{norm}} = |E|^2 / \max(|E|^2)$ over the same longitudinal cross-section and therefore illustrates the spatial localisation and single-lobe character of the cavity mode. Because each field is normalised to its own maximum, the colour scale is used to compare mode shape rather than absolute field strength.

5.5 Spectral and Modelled Single-Photon Performance

The EM results describe how strongly each micropillar enhances and redirects emission, but not how good the emitted photons are. Two properties matter here and arise from different physics, so they must be assessed separately: indistinguishability, meaning how closely photons emitted in separate excitation cycles resemble one another, and photon-number purity, meaning how effectively extra photons are suppressed within a single cycle. Coupling between the QD and lattice vibrations splits the emission into a narrow ZPL and a broader PSB. The single-photon source model assigns 90% of the intrinsic spectral weight to the ZPL and 10% to the PSB [6, 18], then weights that spectrum by the simulated wavelength-dependent Purcell response and the top-channel fraction ξ. Everything following this section of the chapter is therefore model derived from the EM results, not a direct FDTD output.

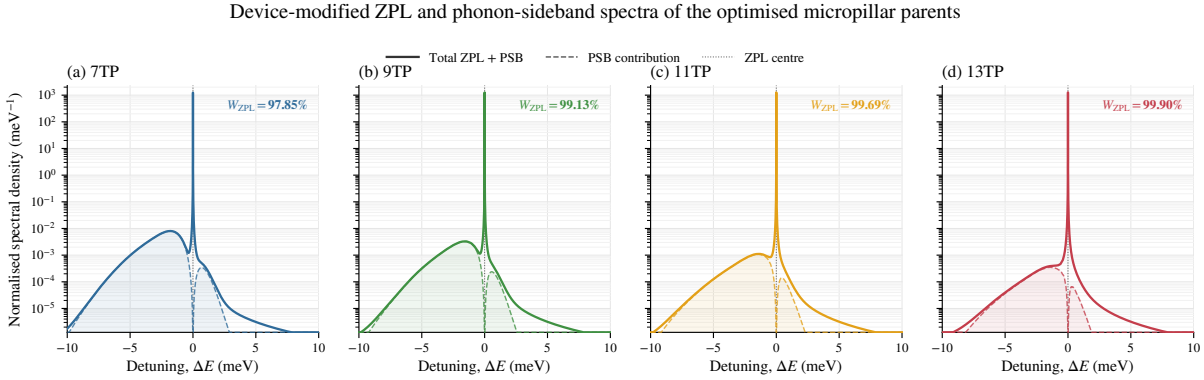


Figure 5.5. Device-modified ZPL and PSB spectra for the optimised 7TP, 9TP, 11TP and 13TP micropillar designs. The solid curve shows the total modelled spectrum and the dashed curve the PSB contribution. W_{ZPL} gives the integrated fraction of useful top-channel emission contained within the ZPL. The spectra are plotted relative to the ZPL centre and normalised to compare spectral composition rather than absolute emitted power.

The ZPL is concentrated at the optical transition while the PSB spans a much wider energy range, so a resonance that narrows and strengthens increasingly favours ZPL emission and leaves more of the sideband outside the strong cavity response. Consequently, the top-channel ZPL fraction rises from 97.85% for 7TP to 99.90% for 13TP, cutting the PSB contribution from 2.15% to 0.10%. The shrinking absolute gains at high TP follow from the bound at 100% and are not evidence of saturation: from 11TP to 13TP W_{ZPL} adds only 0.21 percentage points, yet the residual PSB fraction falls threefold, from 0.31% to 0.10%.

Section 3.3.2 estimates indistinguishability from the top-channel ZPL fraction, assuming phonon scattering dominates distinguishability [6, 25]. It rises from 95.75% for 7TP to 99.80% for 13TP. These are idealised predictions rather than measured visibilities: pure dephasing and slow spectral diffusion are not separately included.

Photon-number purity moves in the opposite direction. Section 3.3.3 models the impact of finite-pulse re-excitation using the short-pulse approximation of Hanschke et al. [17]. If the dot emits a photon and returns

to its ground state while the pulse is still on, the remaining portion of that pulse can excite it again and cause a second photon to be emitted in the same pulse cycle. Consequently, the relationship between pulse duration and radiative lifetime becomes significant.

Family	W_{ZPL} (%)	Modelled I (%)	T (ps)	$g^{(2)}(0) \times 10^3$
7TP	97.85	95.75	0.50	0.39
9TP	99.13	98.27	0.94	1.23
11TP	99.69	99.38	1.75	4.06
13TP	99.90	99.80	3.20	12.89

Table 5.4. Model-derived spectral and photon-quality metrics for the final micropillar designs. W_{ZPL} is obtained from the phonon-weighted top-channel spectrum of Section 3.3.1; I follows the approximation defined in Section 3.3.2. T is the bandwidth-compatible pulse duration calculated from the fitted Purcell linewidth, and $g^{(2)}(0)$ is the Hanschke two-level-system estimate defined in Section 3.3.3. Lower $g^{(2)}(0)$ indicates stronger suppression of multiphoton emission.

Table 5.4 displays why that ratio worsens with TP count. A narrower cavity linewidth lengthens the shortest bandwidth-compatible excitation pulse, from 0.50 ps at 7TP to 3.20 ps at 13TP, while the stronger Purcell enhancement shortens the radiative lifetime. Both timescales move in the same direction for re-excitation, and the modelled $g^{(2)}(0)$ rises accordingly from 3.90×10^{-4} to 1.29×10^{-2}. Stronger confinement thus enables better ZPL selectivity and modelled indistinguishability with a quantifiable trade-off in photon-number purity

5.6 Engineering Interpretation & Comparison Basis

None of the four families discussed in this section are optimal across all metrics. Across them, adding upper-DBR pairs improves some source properties and degrades the others, resulting in a trade-off with two clear ends. One of these ends belongs to the 7TP, with the broadest resonance, lowest modelled $g^{(2)}(0)$ and the strongest photon-number purity. At the other end, the 13TP is the strongest on peak Purcell enhancement, internal useful upward fraction, spectral purification and modelled indistinguishability. The same confinement which produces those results consequently gives it the narrowest resonance and the largest re-excitation penalty. The 9TP and the 11TP lie between the other two, shifting gradually from one set of strengths to the other, with neither constituting a distinct operating regime.

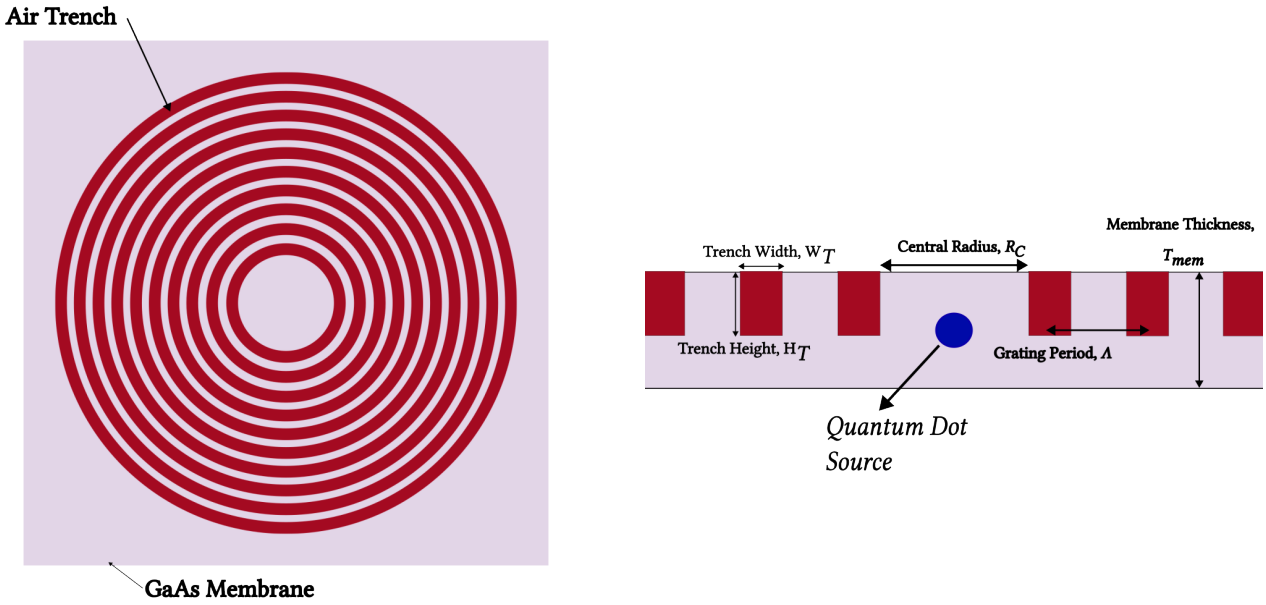
No single metric settles the choice. Which design should be chosen depends on which properties matter most for the intended use. Rapid emission, high useful upward output and strong ZPL selectivity all point to the high-TP end. By contrast, multiphoton suppression and broader spectral operation point the other way. One caveat applies to all four. The resonance is still situated near 1295 nm, and reaching the design target would require a further round of tuning.

What this chapter establishes is a self-contained quantiative reference for the micropillar architecture. Chapter 7's hybrid was developed independently. Nothing in it derives from this optimisation, and no design relationship should be implied from the two appearing in the same report. Its focus on the 13TP regime has a specific and separate justification to the reported conclusions within this chapter. Three properties motivate this. A hybrid architecture is build for enhancement, useful upward output and spectral purification, and amongst the four families here, the 13TP is the strongest on each. This is not a verdict on the 7TP, as it remains the relevant reference wherever photon-number purity is the binding requirement. Niether end comes free, and the chapter's contribution is to make the price of each explicit.

6. Circular Bragg Grating (CBG) - MM

6.1 Design Background, Architecture and Objectives

The CBG consists of a central GaAs disk containing the QD dipole, surrounded by concentric etched trenches. Its operating principle was introduced in Chapter 2; here the focus is on how the central radius R_c, grating period Λ, trench width w, trench depth d, membrane thickness t and number of trenches N control resonance, Purcell enhancement and directional power redistribution. The QD is represented by an in-plane dipole at the bullseye centre.



(a) Top view.

(b) Cross-sectional view.

Figure 6.1. Circular Bragg grating geometry and principal design parameters.

The associated geometric quantities are

$$w_{\text{ridge}} = \Lambda - w, \quad r_{\text{etch}} = \frac{d}{t}, \quad t_{\text{rem}} = t - d. \tag{6.1}$$

A fully etched grating has $d = t$, $r_{\text{etch}} = 1$ and $t_{\text{rem}} = 0$.
Design development proceeded in three stages. First, the 948.9 nm CBG of Davanço *et al.* [35] was reconstructed to validate the model. The geometry was then translated to 1300 nm while retaining comparable performance. Finally, a DBR-assisted "Green Grass" design and a fully etched air-gap "Blue Sky" variant were explored. Candidate designs were assessed using resonance position, F_P, Q and directional power fractions rather than any single metric.

6.2 Reference Design and Validation

The numerical workflow was benchmarked against the approximately 948.9 nm CBG reported by Davanço *et al.* [35]. The reference structure is a 190 nm suspended GaAs membrane with ten partially etched trenches and a centred in-plane dipole. The principal geometric parameters are listed in Table 6.1.
Using the common extraction procedure of Chapter 3, the reconstructed model gave a peak $F_P = 11.85$ at 949.51 nm with $\Delta\lambda_{\text{FWHM}} = 3.73$ nm and $Q = 254.66$. The resonance differs from the published value by only 0.61 nm, while the enhancement remains of the same magnitude. The reconstructed directional fractions were 80.43% upward, 17.03% downward and 2.54% lateral. These upward values are not equated to the finite-NA collection efficiency reported in the literature because the monitor calculation includes all power crossing the upper boundary

Table 6.1. Reference CBG geometry used for numerical validation.

Parameter	Symbol	Value
Membrane thickness	t	190 nm
Number of trenches	N	10
Grating period	Λ	360 nm
Central radius	R_c	720 nm
Trench width	w	110 nm
Trench depth	d	$\approx$ 133 nm
Dipole position	(x, y, z)	$(0, 0, 0)$
Dipole orientation	—	In-plane

Table 6.2. Metric comparison to [35]

Metric	Published	Reconstructed
λ_{res}	948.9 nm	949.51 nm
Peak F_P	$\approx$ 11	11.85
FWHM	$\approx$ 5 nm	3.73 nm
Q	$\approx$ 200	254.66
Upward emission	NDR	80.43%
Downward emission	NDR	17.03%
Lateral emission	NDR	2.54%

NDR: not directly reported.

6.3 Wavelength Translation and Geometric Sensitivity

The validated geometry was first scaled by $S = 1300/948.9 \approx 1.370$. This preserves the principal dimensionless geometric ratios and therefore provides a physically motivated initial estimate, although material dispersion prevents exact wavelength scaling.

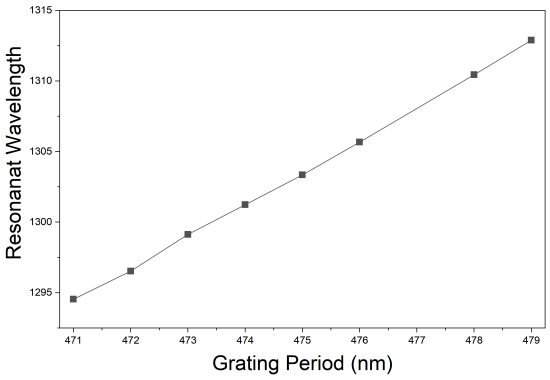
Table 6.3. Uniform scaling of the validated CBG towards 1300 nm.

Design	t	Λ	R_c	w	d	$\Lambda - w$	d/t	N
948.9 nm	190 nm	360 nm	720 nm	110 nm	133 nm	250 nm	≈ 0.70	10
Scaled 1300 nm	260.3 nm	493.2 nm	986.4 nm	150.7 nm	182.2 nm	342.5 nm	≈ 0.70	10

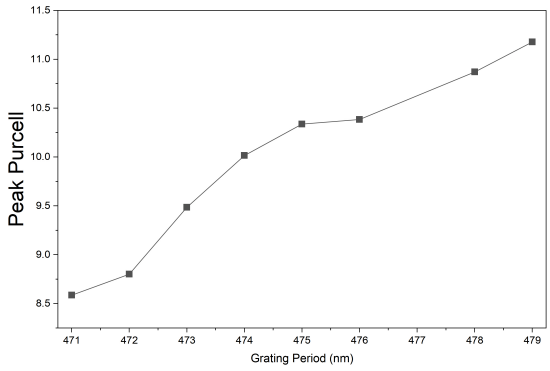
The strongest wavelength control was obtained from Λ and R_c. For a radial propagation constant $\beta_r \simeq 2\pi n_{\text{eff}}/\lambda$, the relevant phase conditions may be written approximately as

$$\beta_r \Lambda \simeq 2\pi m, \qquad 2\beta_r R_c + \phi_g \simeq 2\pi q. \tag{6.2}$$

Thus, Λ controls the phase accumulated between successive grating features, while R_c controls the optical path from the emitter region to the first trench. Figure 6.2 shows the strong dependence of resonance wavelength on Λ, while retaining a comparatively strong Purcell response. The trench-width and trench-depth sweep in Figure 6.3 produced weaker wavelength control and generally reduced the peak Purcell factor for combinations approaching 1300 nm.

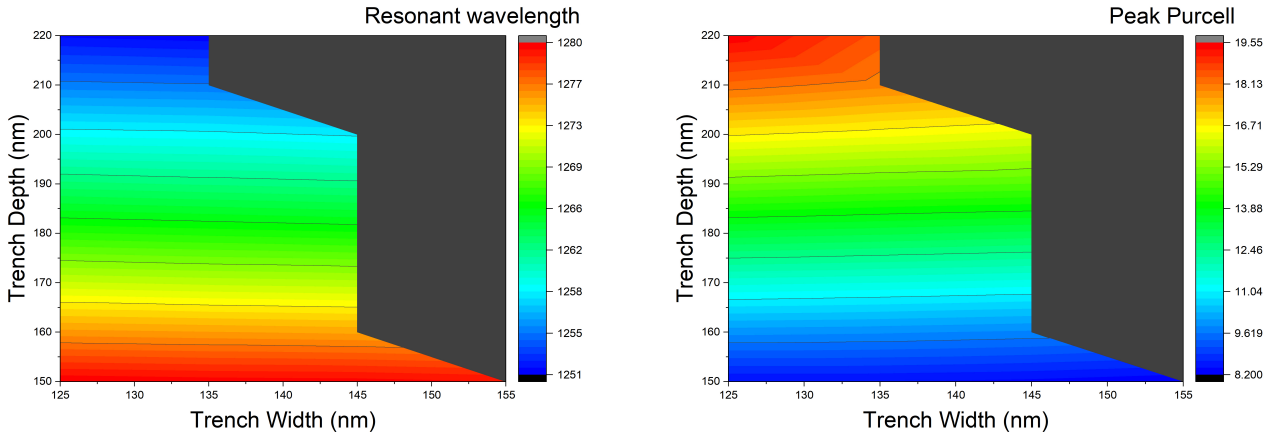


(a) Resonance wavelength.



(b) Peak Purcell factor.

Figure 6.2. Effect of grating period Λ on the resonance wavelength and peak Purcell factor during wavelength translation.



(a) Resonance wavelength.

(b) Peak Purcell factor.

Figure 6.3. Effect of trench width w and trench depth d on resonance wavelength and peak Purcell factor. Compared with Λ, the trench dimensions provide weaker spectral tuning over the investigated range.

The broad sweep placed the 1300 nm operating region near $\Lambda = 470$–480 nm, after which local refinement produced the baseline summarised in Table 6.4. The purpose of this stage was spectral translation rather than optimisation.

Table 6.4. Reference CBG and wavelength-translated 1300 nm baseline.

Design	λ_{res}	Peak F_P	Q	Upward	Downward	Lateral
Reference	949.51 nm	11.85	254.66	80.43%	17.03%	2.54%
1300 nm baseline	1300.13 nm	12.68	231.50	81.30%	17.08%	1.62%

6.4 DBR-Assisted Design and Optimisation

The dominant remaining loss in the translated baseline was the 17.08% downward channel. A GaAs/AlAs DBR was therefore introduced beneath the membrane to suppress substrate-directed radiation.

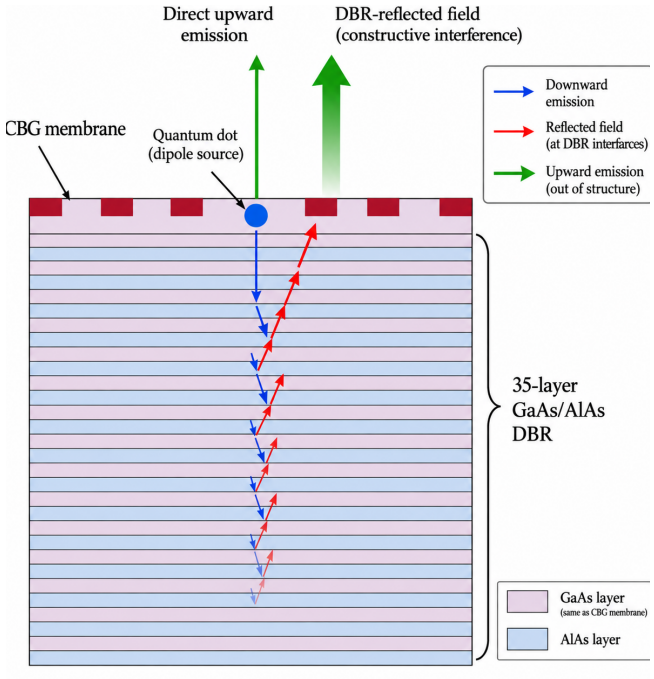


Figure 6.4. Initial DBR-assisted CBG and reflected-field mechanism.

With 35 layers, downward emission fell from 17.08% to 0.046% and upward emission increased from 81.30% to 86.43%. The suppressed power was not transferred exclusively upward, however, since lateral emission rose from 1.62% to 13.53%. The reflector also shifted the principal resonance from 1300.13 to 1268.63 nm and reduced F_P from 12.68 to 10.57 (Q : 231.5 → 194.0). The geometry therefore required retuning. A grating-period sweep was first performed at fixed R_c to restore the resonance. Figure 6.5 shows that increasing Λ simultaneously reduced Q and F_P while increasing upward efficiency. This demonstrates the central design trade-off: stronger useful out-coupling is itself a radiative loss channel, so a lower-Q resonance can still provide a larger upward fraction.

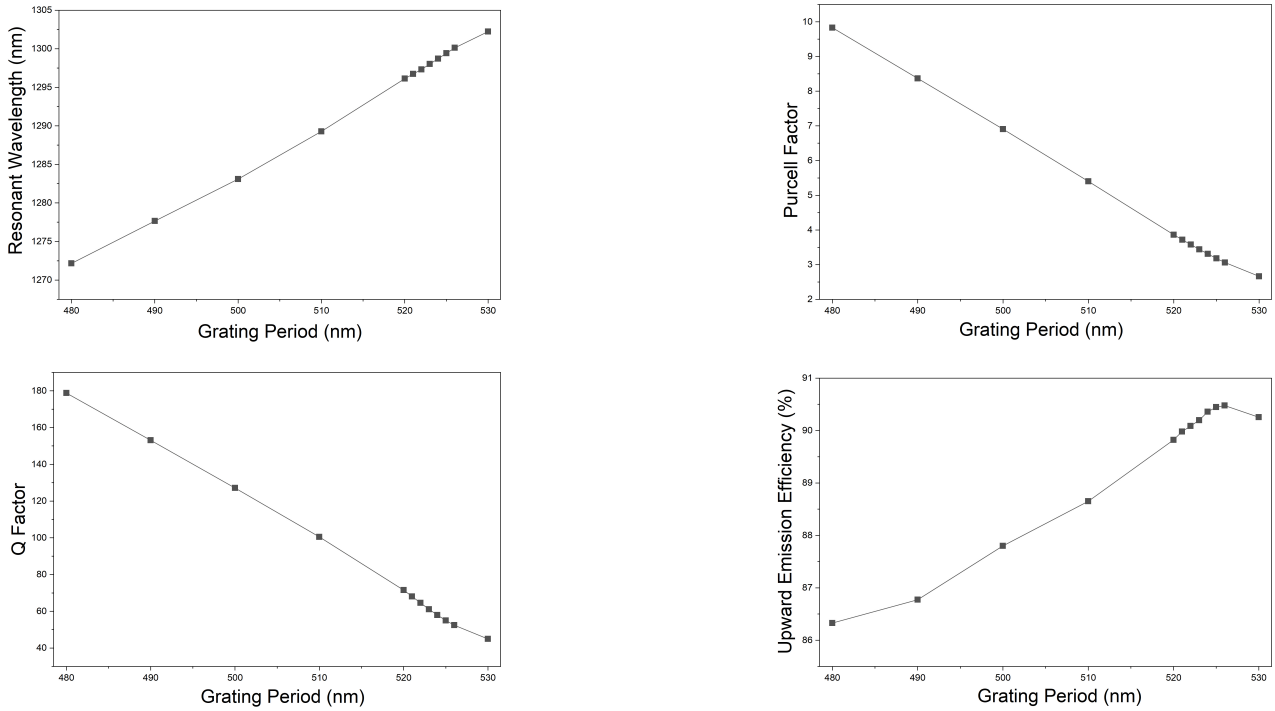


Figure 6.5. Effect of grating period on the DBR-assisted CBG.

After spectral alignment, w and d were swept to test whether the trench cross-section offered comparable leverage. Their effect on F_P was weak over the investigated range. In contrast, allowing Λ and R_c to vary independently produced much stronger changes in both resonance position and peak enhancement, confirming these radial dimensions as the more appropriate parameters to tune

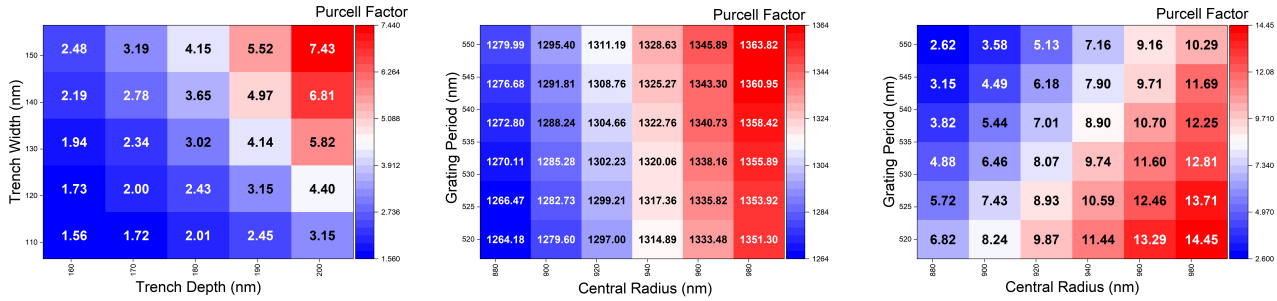
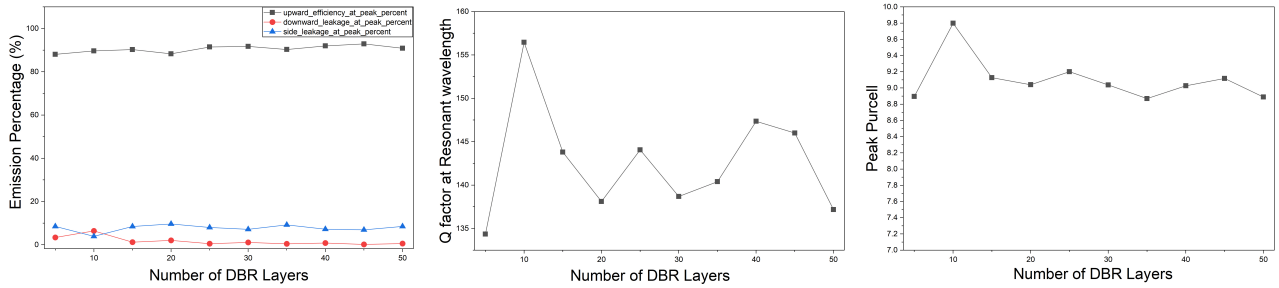


Figure 6.6. Two-dimensional geometry sweeps. Trench dimensions provide comparatively weak tuning, whereas Λ and R_c strongly control the radial resonance.

Reflector thickness was then swept. Above approximately 25 layers, upward efficiency changed only modestly while downward leakage was already typically below 1%; no systematic improvement in Q or F_P was observed. A 25-layer DBR was therefore retained as the point of diminishing optical return. The 10-layer case produced local maxima in Q and F_P, but also the largest downward leakage (~ 6.4%), so it was treated as a local phase-sensitive interference feature rather than a separate optimum.



(a) Upward efficiency. (b) Q. (c) Peak F_P.

Figure 6.7. Dependence on DBR layer number.

Table 6.5. Comparison of the 1300 nm baseline and selected Green Grass design.

Design	t	Λ	R_c	w	d	N	DBR layers	λ_{res}	Peak F_P	Q	Upward	Downward	Lateral
1300 nm baseline	260.3 nm	474 nm	948 nm	135 nm	180.5 nm	10	0	1300.13 nm	12.68	231.50	81.30%	17.08%	1.62%
Green Grass	260.3 nm	526.0 nm	924.0 nm	151.5 nm	200.5 nm	10	25	1300.330 nm	9.023	143.57	94.27%	0.15%	5.58%

The Green Grass design therefore improves directionality rather than resonant confinement: downward emission is almost eliminated and upward emission rises to 94.27%, while both F_P and Q decrease relative to the translated baseline.

6.5 Blue-Sky Design

Although the Green Grass design achieved strong upward directionality, its resonant enhancement remained comparatively modest, with $F_P = 9.023$ and $Q = 143.57$. The preceding trench-depth sweeps showed that F_P continued to increase as d approached the membrane thickness, motivating investigation of the fully etched limit, $d = t$.

Fully etching the membrane removes the residual GaAs beneath each trench and maximises the refractive-index modulation between the GaAs ridges and surrounding air. This can strengthen radial confinement and modify the available radiative leakage channels. In terms of the Purcell relation introduced in Eq. (2.6), stronger confinement may increase Q and potentially reduce the effective mode volume V_{eff}, both of which favour increased Purcell enhancement.

A second degree of freedom was introduced by separating the membrane from the DBR with an air gap g. The DBR behaviour observed in Section 5.4 showed a strong dependence on reflected-field interference. Changing g modifies the propagation phase accumulated by the field between the membrane and reflector, allowing the returning field to interfere differently with the CBG mode. The fully etched geometry and air-gap separation were therefore investigated together as a new design regime.

An initial simulation was first performed by applying the fully etched geometry and air-gap separation to the Green Grass design without retuning the radial grating dimensions. This provided a direct indication of how strongly the new configuration altered the electromagnetic response before further optimisation. The result is compared with Green Grass in Table 6.6.

Table 6.6. Comparison of the Green Grass design with the initial fully etched air-gap configuration.

Design	λ_{res} (nm)	F_P	Q	FWHM (nm)	Upward (%)	Downward (%)	Lateral (%)
Green Grass	1300.33	9.023	143.57	~ 9.06	94.27	0.15	5.58
Fully etched + air gap	1256.99	27.650	392.55	3.20	94.33	0.29	5.38

The immediate response was substantial. Peak F_P increased from 9.023 to 27.650 and Q increased from 143.57 to 392.55, while the upward power fraction remained essentially unchanged at approximately 94%. The modification therefore produced considerably stronger resonant enhancement without sacrificing the directionality obtained in the Green Grass design.

However, the resonance shifted from 1300.33 nm to 1256.99 nm. This initial result was therefore not itself an acceptable O-band design, but demonstrated that the fully etched air-gap configuration contained a significantly higher-performance regime worth optimising.
Having established the strong response of the modified geometry, the air gap g and trench depth d were swept together to determine how the two parameters controlled the enhanced regime. Figure **??** shows the resulting peak Purcell factor and quality factor.

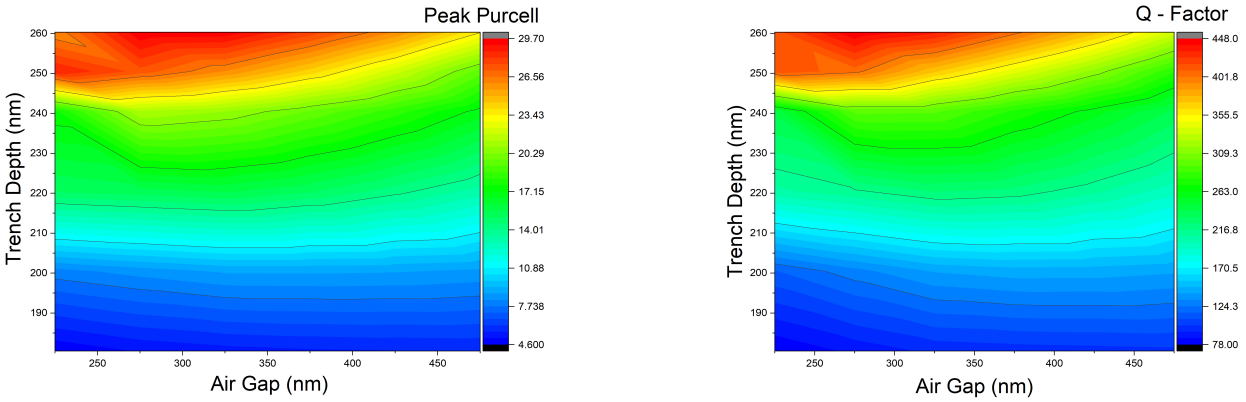


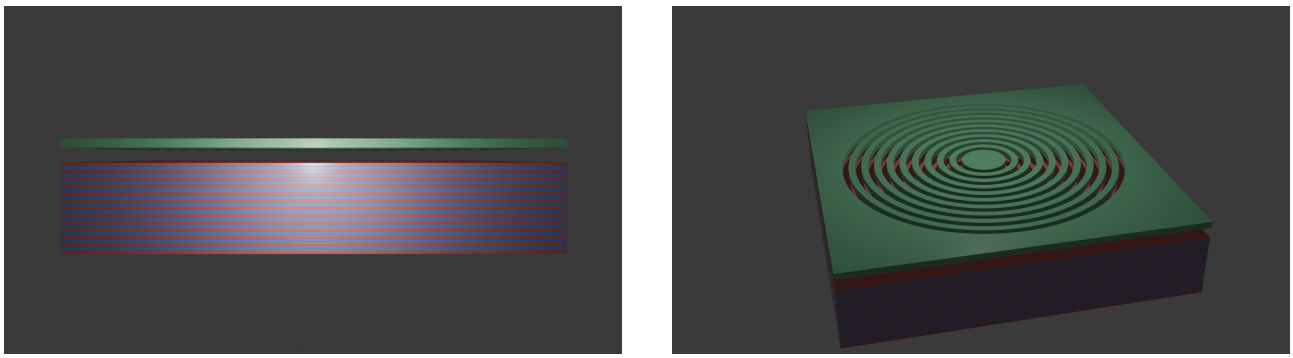
Figure 6.8. Combined air-gap and trench-depth sweep, showing the dependence of (a) peak Purcell enhancement and (b) cavity quality factor on air gap g and trench depth d.

Figure 6.8 shows that Q and F_P follow closely correlated trends across the investigated parameter space. Trench depth provides the dominant variation: both quantities increase strongly as d approaches the fully etched condition $d = t$. The air gap produces a weaker, but still observable, modulation consistent with its role in controlling the phase of the DBR-reflected field.
The correlation between Q and F_P is consistent with the Purcell scaling discussed in Eq. (2.6). The results therefore indicate that the principal improvement is associated with increased resonant confinement in the fully etched regime, with the air gap providing an additional phase-sensitive optimisation degree of freedom.
The modification produced a substantial increase in resonant enhancement. The peak Purcell factor increased from 9.02 to 27.65, while the quality factor increased from 143.6 to 392.5. Importantly, this stronger resonant response was obtained without a corresponding loss of directionality: the upward power fraction remained effectively unchanged at 94.33%, compared with 94.27% for Green Grass, while the lateral contribution decreased slightly from 5.58% to 5.38%.
The coupled g–d sweep therefore identified a regime with substantially improved Q and F_P while preserving strong upward emission. At this stage, the principal remaining requirement was spectral rather than electromagnetic: the resonance had shifted away from the target 1300 nm wavelength.
section 6.3 established Λ and R_c as the dominant controls of the CBG resonance. These parameters were therefore retuned while retaining the fully etched membrane and selected air-gap condition, allowing the resonance to be restored without abandoning the enhanced confinement regime identified above.
The resulting geometry is summarised in Table 6.7. The final design reached $\lambda_{res} = 1300.015$ nm with $F_P = 30.872$, $Q = 415.95$ and an upward power fraction of 97.01%.

Design	t	Λ	R_c	w	d	d/t	N	**DBR layers**	g	λ_{res}	F_P at 1300 nm	Q	**Upward efficiency**
Blue Sky	260.3 nm	538.0 nm	946.0 nm	151.5 nm	260.3 nm	1.00	10	25	425 nm	1300.015 nm	30.872	415.95	97.01%

Table 6.7. Selected Blue Sky CBG geometry and performance.

The Blue Sky design therefore recovers substantially stronger resonant enhancement while retaining high upward extraction. It is treated as a separate design regime rather than a simple continuation of Green Grass because both the etch topology and membrane–DBR phase condition have changed.



(a) Air-gap geometry.

(b) Fully etched geometry.

Figure 6.9. A 3D model of the final BLue-Sky design, presenting the air gap in (a) and the fully etched rings through the membrane (b).

6.6 Spectral and Source-Level Performance

The electromagnetic optimisation of sections 6.3 to 6.5 establishes the photonic response of the three CBG designs, but does not alone establish their performance as single-photon sources. The wavelength-translated baseline, Green Grass and Blue Sky designs were therefore evaluated using the common metric extraction and source-level processing framework defined in Sections 3.2 and 3.3. This retains the distinction between quantities obtained directly from the FDTD simulations and those derived from the subsequent source model. The resulting comparison is summarised in Table 6.8.

Metric	**1300 nm baseline**	**Green Grass**	**Blue Sky**
Electromagnetic quantities			
Peak Purcell factor, F_{P}	12.68	9.023	30.872
Quality factor, Q	231.50	143.57	415.95
Purcell FWHM (nm)	5.62	9.06	3.13
Active coupling, β	98.38%	94.42%	97.44%
Passive efficiency, η	82.64%	99.85%	98.75%
Internal efficiency, ξ	81.30%	94.27%	97.01%
FOM			
Effective enhancement, F_{eff}	11.69	8.58	25.37
Top-channel ZPL fraction	95.60%	95.4%	98.28%
Top-channel PSB fraction	4.40%	4.6%	1.72%
$g^{(2)}(0)$	3.2×10^{-3}	1.3×10^{-3}	1.18×10^{-2}
Indistinguishability, $\mathcal{I}$	91.40%	92.2%	96.59%
Device efficiency, ϵ	81.64%	94.2%	96.0%

Table 6.8. Electromagnetic and FOM comparison of the three established CBG designs. All values are calculated using processes stated in Chapter 3

The directional behaviour differs clearly between the three CBG regimes. For the wavelength-translated baseline, $\beta = 98.38\%$, $\eta = 82.64\%$ and $\xi = 81.30\%$, with the lower η caused by the 17.08% downward-emission channel. Introducing the DBR in Green Grass reduces downward emission to 0.15%, increasing η to 99.84% and ξ to 94.27%, despite a slight reduction in β to 94.42% from increased lateral emission. Green Grass therefore primarily improves directionality, while Blue Sky further increases the useful upward fraction to 97.01% while recovering stronger resonant enhancement.

The spectral response shows a different trade-off. Green Grass has the weakest and broadest resonance, with $F_{\mathrm{P}} = 9.023$, $Q = 143.57$ and a FWHM of approximately 9.06 nm, whereas Blue Sky reaches $F_{\mathrm{P}} = 30.872$ and $Q = 415.95$ with a narrower 3.13 nm linewidth. The stronger spectral selectivity of Blue Sky favours greater ZPL concentration and PSB suppression, while the broader Green Grass response provides greater spectral tolerance.

The source-level results reinforce this trade-off. Green Grass gives the lowest calculated multiphoton contribution, $g^{(2)}(0) \approx 1.3 \times 10^{-3}$, compared with 3.2×10^{-3} for the baseline and approximately 1.18×10^{-2} for Blue Sky. Blue Sky therefore favours stronger enhancement, faster emission and spectral selectivity, while Green Grass favours photon-number purity and robust directionality. Neither design is universally superior; the preferred CBG depends on the relative importance of purity, bandwidth, enhancement and useful output.

6.7 Conclusions and Limitations

The CBG study identified both high-performing designs and the physical parameters governing their behaviour. Validation against the 948.9 nm reference provided confidence in the numerical model before translation to the O-band. Across the sweeps, Λ and R_c were the dominant controls of resonance wavelength, while DBR layer number primarily reduced downward leakage, with diminishing returns beyond approximately 25 layers. In the strongly etched regime, trench depth d had the greatest influence on F_P and Q, while the membrane–DBR air gap g provided a weaker phase-sensitive adjustment. Trench width was comparatively less influential over the ranges studied.

These trends produced two distinct optimised regimes. Green Grass uses the DBR primarily to suppress downward emission, increasing ξ from 81.30% to 94.27% while accepting lower F_P and Q. Blue Sky combines full etching with membrane–DBR phase control, reaching $F_\mathrm{P} = 30.872$, $Q = 415.95$ and 97.01% upward emission. However, its stronger spectral selectivity is accompanied by a shorter radiative timescale and a larger finite-pulse multiphoton contribution. The preferred design therefore depends on application priorities rather than peak Purcell enhancement alone.

Blue Sky should nevertheless be regarded as an idealised numerical design rather than a fabrication-ready structure. It assumes a perfectly fully etched membrane, a controlled 425 nm air gap, ideal geometry and a centred, correctly oriented QD, while fabrication tolerances, mechanical supports and emitter-position uncertainty were not included. Furthermore, the reported upward efficiencies represent power crossing the simulation boundary rather than finite-NA or fibre-coupled collection. Future work should therefore examine fabrication tolerance, QD position and orientation, realistic support structures and finite-NA collection, together with simultaneous multi-parameter optimisation. Overall, the study shows that resonance, confinement and directionality are controlled by different geometric parameters, and that identifying these dependencies is central to effective CBG optimisation.

6.7.1 Mesh-Order Verification

As discussed in Sections 3.1.1 and 3.1.4, increasing the FDTD mesh order increases the spatial resolution of the Yee-cell grid, improving numerical precision at greater computational cost. To assess whether the M3 resolution used for the principal CBG parameter sweeps was sufficient, the final Green Grass and Blue Sky geometries were re-simulated at mesh order 5 (M5).

For Green Grass, the resonance shifted from 1300.33 nm to 1296.13 nm, while the peak Purcell factor increased from 9.02 to 9.68 and Q from 143.57 to 155.81. Despite these changes, the upward power fraction remained effectively unchanged, increasing only from 94.27% to 94.40%.

For Blue Sky, the resonance shifted more modestly from 1300.02 nm to 1298.73 nm. The peak Purcell factor increased from 30.87 to 32.80, while Q increased from 415.95 to 430.10. The upward fraction likewise remained stable, changing from 97.01% to 97.31%.

The higher-resolution simulations therefore preserve the principal comparative conclusions of the M3 study: Green Grass remains primarily a high-directionality design, while Blue Sky retains substantially stronger resonant enhancement together with high upward emission. The larger mesh-dependent variation in resonance wavelength indicates that absolute spectral position is more sensitive to discretisation than the overall performance ranking. This behaviour is considered further alongside the other architectures in the cross-architecture mesh-order verification of Section 9.2.1, and supports the use of M3 for the computationally intensive parameter sweeps while retaining appropriate caution regarding absolute spectral position.

7. Bragg Ring Micropillar - CB

7.1 Background & Architecture

The design of a Bragg Ring Micropillar (BRM) involves the placement of n concentric rings around a central micropillar. This mechanism is closely related to the operating principles of a CBG (Chapter 6), in which laterally propagating modes are re-directed towards the device's centre by successive ring structures separated by air gaps. In the BRM case, while the central pillar retains the vertically resonant cavity through its top-and-bottom DBR configuration (Chapter 5), the addition of concentric rings, in theory, would act to further suppress a significant source of remaining unwanted leakage, namely the side channel ($\Gamma side$).

The BRM structure itself, commonly dubbed a 'hybrid architecture', has been the subject of increased theoretical and experimental work in recent literature. An early numerical investigation of a related device was presented by Blokhin et al. [36], who modelled the effects of placing a lateral CBG structure around a micropillar device. Results obtained through 3D-FDTD modelling presented a photon extraction efficiency of approximately **83%** into the upper hemisphere. More recently, simulations performed on a micropillar cavity surrounded by two concentric rings predicted an approximately **4**× of collection efficiency directly attributed to the suppression of lateral emission into undesirable background radiation modes [37]. Experimentally, however, an increase of only **1.3**× was observed, attributed speculatively by the authors to losses related to deviations of the fabricated device from the canonical best-performing geometry. This discrepancy highlights

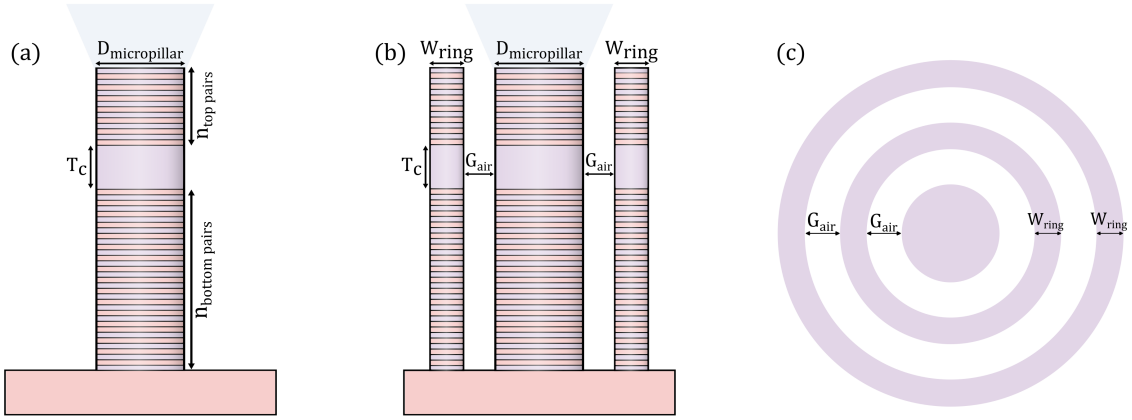


Figure 7.1. Illustrative diagrams of micropillar and BRM geometry. **(a)** bare micropillar, **(b)** micropillar-BRM with one concentric ring (cross section), **(c)** top-down view of micropillar-BRM with two concentric rings of equal width and spacing. G_{air} denotes the air gap between the pillar and concentric rings, W_{ring} denotes the width of the aforementioned rings, T_C is the thickness (in the Z direction) of the central cavity in which the QD is housed. $D_{micropillar}$ is the diameter of the central pillar, and $n_{bottompairs,toppairs}$ are the number of DBR pairs on the bottom and top DBR stacks respectively.

strong geometric variances of the BRM architecture and, as such, motivates the search for high-performing geometric regimes.

7.2 Design Motivation & Search Space

In the final architecture, the concentrically-placed annuli reproduce the vertical epitaxial structure of the central micropillar, which comprises of a 35.5-pair bottom DBR stack ($N_{bottom} = 71$), a Gallium Arsenide ($GaAs$) central cavity of $T_C = 0.381\mu m$, and a 17-pair top DBR stack ($N_{top} = 34$). The central pillar design reproduces an established baseline geometry presented by Dlaka et al. [6] The DBR stacks comprise of alternating layers of $GaAs$ and Aluminium Arsenide ($AlAs$) *(Figure 7.2)*. This was preferred over a separate homogeneous structure to preserve a monolithic architecture in which the annuli retain the refractive-index modulation of the central pillar. This approach is consistent with recently demonstrated pillar-ring architectures, in which the pillar and the annuli are created from the same epitaxial structure [10, 37]. The potential benefits from this approach are two-fold; in addition to preserving the material properties of the central pillar, the structure

becomes more practically fabricable as the central pillar and n number of concentric annuli can originate from one epitaxially grown wafer, with etching techniques used to define parameters W_{ring} and G_{air}.

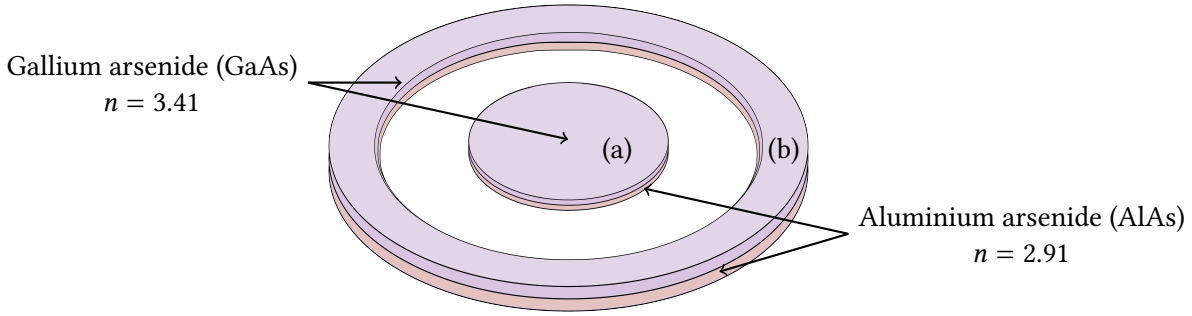


Figure 7.2. Illustration of a DBR pair with materials and their respective refractive indices annotated. (a) central micropillar, (b) First concentric ring

Given the demonstrated dependence of the EM field profile on both W_{ring} and G_{air} [10, 37], a two-dimensional (2D) search was required to thoroughly investigate the mutually dependent effect of concentric-ring thickness and separation on the field distribution. Given a target resonance of $\lambda_0 = 1300\,\text{nm}$, the corresponding quarter-wave dimensions provide useful first-order estimates of $G_{\text{air}} \approx \lambda_0/4 \approx 325\,\text{nm}$ and $W_{\text{ring}} \approx \lambda_0/(4n_{\text{GaAs}}) \approx 95\,\text{nm}$, where $n_{\text{GaAs}} \approx 3.41$ is the refractive index of GaAs at the target wavelength. The latter is used only as an approximate characteristic semiconductor index, since the annuli reproduce the vertically layered GaAs/AlAs epitaxial stack rather than a homogeneous GaAs region. One (1R) and two ring (2R) configurations were selected as the foundation for optimisation, allowing for the effect of introducing an additional concentric ring to be investigated while retaining an identical central pillar geometry.

An initial 7×7 geometry map was conceived for both the 1R & 2R case, providing 49 different geometries for each ring configuration. The air gap was varied from 200nm to 440 nm in 40 nm increments, while the ring radial width was varied from 80 to 140 nm in 10 nm increments. These different physical step sizes provide the same seven-point sampling along each parameter axis while also giving broadly comparable changes in optical path length. The corresponding path changes are

$$\Delta\text{OPL}_{\text{air}} = n_{\text{air}}\Delta G_{\text{air}} \approx 40\,\text{nm} \qquad \Delta\text{OPL}_{\text{ring}} = n_{\text{GaAs}}\Delta W_{\text{ring}} \approx 34.1\,\text{nm}. \tag{7.1}$$

And since accumulated phase scales with $n\Delta d$, the corresponding phase increments at $\lambda_0 = 1300\,\text{nm}$ are

$$\Delta\phi_{\text{air}} = \frac{2\pi}{\lambda_0}\Delta\text{OPL}_{\text{air}} \approx 0.193\,\text{rad} \qquad \Delta\phi_{\text{ring}} = \frac{2\pi}{\lambda_0}\Delta\text{OPL}_{\text{ring}} \approx 0.165\,\text{rad}. \tag{7.2}$$

Therefore the two axes, despite a large difference in increment size, are sampled at a similar phase resolution.

7.3 One Ring

The first stage of the investigation established the effect of adding one concentric ring around the central pillar. The one-ring sweep incrementally varied the ring radial width, W_{ring}, and the pillar-to-ring air gap, G_{air}. The results produced a dependence of lateral emission on both the ring-width and pillar-to-ring air gap.

As demonstrated in Figure 7.3, the strongest geometric dependence is seen by the lateral channel, which varies by more than a factor of two across the parameter space. In contrast, the peak Purcell factor (a) and top transmission (b) are significantly less variable. As seen in Figure 7.4, this reduction in lateral transmission naturally increases the active coupling factor (β) since less of the overall radiation is lost through the side channel. This does not, however, map cleanly to increased top extraction. In areas of heightened β, the reduction in lateral power is accompanied by a reduction in η, suggesting a larger relative bottom contribution as a result of increased side-suppression. The resulting collection efficiency ($\xi = \beta\eta$) remains broadly stable due to this offset, though does drop slightly in areas of high side suppression.

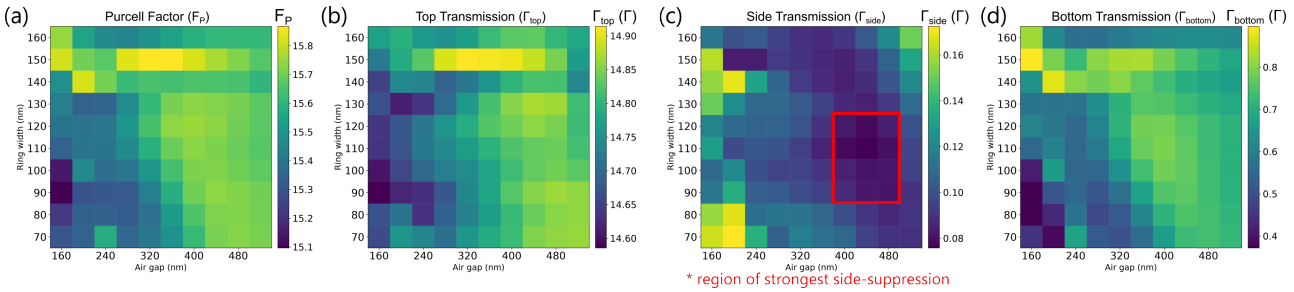


Figure 7.3. One-ring two-dimensional parameter sweep showing the dependence of device performance on ring radial width and pillar-to-ring air gap. Heat maps show (a) Purcell factor, (b) top transmission, Γ_{top}, F_P, (c) side transmission, Γ_{side}, and (d) bottom transmission, Γ_{bottom}.

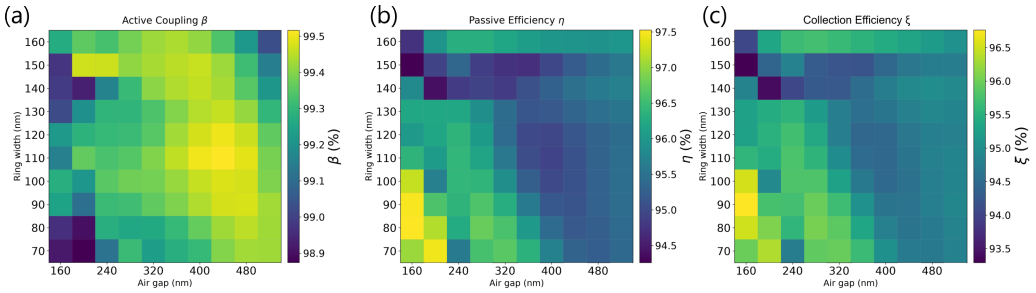


Figure 7.4. One-ring efficiency metrics across the expanded geometric parameter space. (a) active coupling factor, β, (b) passive vertical efficiency, η, and (c) top-extraction efficiency, ξ.

Several localised regions of comparatively high side transmission are present at small G_{air} values, specifically at the upper & lower limits of W_{ring} in Figure 7.3 (c). To investigate further, the geometry map was subsequently extended to 10×10, whilst retaining the existing geometry increments, which determined that these features formed part of broader trends that extended beyond the initial search space. At small values of G_{air}, comparatively high $\Gamma side$ 'islands' are observed around the W_{ring} minima and maxima, with a region of comparatively low $\Gamma side$ between them. This behaviour appears consistent with the phase-sensitive scattering properties of the ring structure, capable of producing periodic patterns of both constructive and destructive interference.

The results of the 1R sweep therefore establish two primary effects. First, the addition of a single concentric ring allows for strong control of the lateral radiation channel while, comparatively, only minimally modifying the Purcell factor, and hence the cavity spontaneous emission rate. Secondly, the additionally-suppressed side emission does not cleanly translate to increased top extraction in all cases, as a large portion of the redirected side emission contributes to the bottom channel. As such, a 1R BRM architecture demonstrates significant potential for side-leakage suppression, but further motivates the search for additional geometric regimes that preferentially direct suppressed leakage into the desired top channel.

7.4 Two Rings

For the two-ring (2R) geometry exploration, both rings were assigned the same radial width and the pillar-to-ring and ring-to-ring air gaps were set to be equal:

$$W_{ring,1} = W_{ring,2}, \qquad G_{air,1} = G_{air,2} \tag{7.3}$$

It is important to note that while this geometry relation is not assumed to be the optimal 2R configuration, but does allow for a controlled search in which the active geometric variables are kept to a minimum. Additionally, the matching of 1R & 2R parameters does, in theory, allow for the response associated with introducing a second concentric ring to be assessed while the central pillar and 1R geometry are held constant. While this approach does not entirely isolate the effect of 2R, since the addition of a second concentric ring modifies the electromagnetic boundary conditions of the overall structure and hence may alter the interference conditions

associated with 1R, it does provide a controlled comparison in which the central pillar and 1R geometry remain fixed.

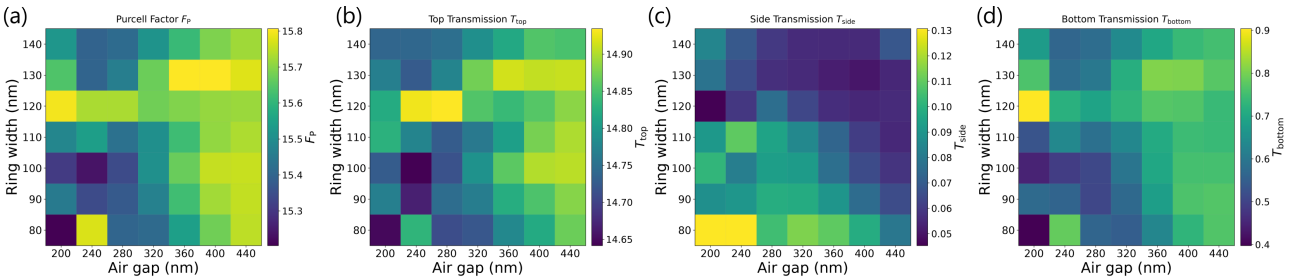


Figure 7.5. Two-ring two-dimensional parameter sweep showing the dependence of device performance on ring radial width and pillar-to-ring air gap. Heat maps show (a) Purcell factor, F_P, (a) top transmission, Γ_{top}, (c) side transmission, Γ_{side}, and (d) bottom transmission, Γ_{bottom}.

The addition of a second concentric ring further reduces side emission throughout a large proportion of the map ($\Gamma_{side,2,min} \geq 0.045\Gamma$ compared to $\Gamma_{side,1,min} \geq 0.075\Gamma$, a decrease of 40%). The lowest side-transmission values of the two-ring sweep are below those obtained using a single ring, demonstrating that the introduction of a second ring can provide additional redirection of the lateral radiation field and further-suppress lateral radiation. Whilst the 2R results demonstrate that stronger lateral suppression is achieved, they do not yet establish whether the additionally suppressed radiation is redirected into the desired top extraction channel. As such, a comparison of the 1R and 2R results is presented in Section 7.5 .

7.5 Comparative Performance of Bare, 1R & 2R Architectures

7.5.1 Additional Lateral Suppression & Redistribution

[note: I still need to discuss ZPL heatmap & mode volume which will take roughly one more page.]
Although 2R provides significant additional suppression of the lateral channel, a reduction of $\Gamma side$ on its own is not sufficient to establish that the additional suppressed power is redirected into the desired top channel mode. As such, additional side suppression was plotted against $\Delta\Gamma top$, suggesting a modest positive correlation between the suppression of unwanted lateral radiation modes and top emitted power.

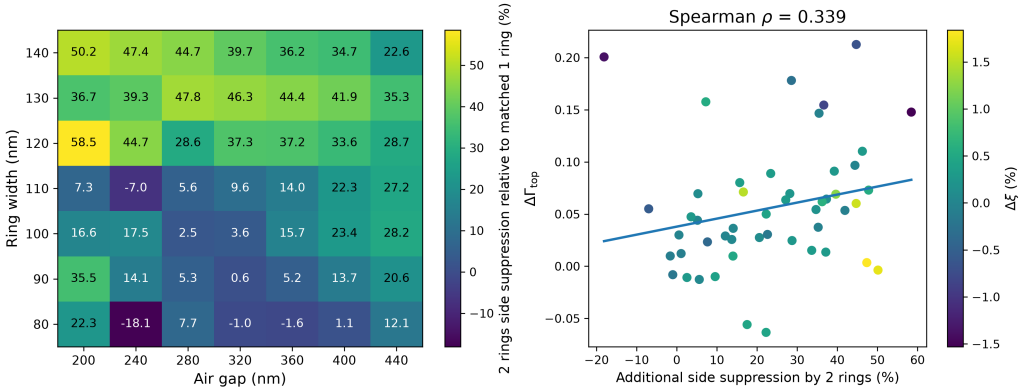


Figure 7.6. Effect of introducing a second concentric ring on lateral suppression and top-channel redistribution. (a) Additional side-emission suppression produced by the 2R relative to the 1R geometry across the investigated parameter space. Positive values indicate reduced Γ_{side}. (b) Relationship between additional side suppression and the change in Γ_{top} ($\Delta\Gamma_{top} = \Gamma_{top,2R} - \Gamma_{top,1R}$), with point colour indicating $\Delta\xi$, ($\Delta\xi = \xi_{2R} - \xi_{1R}$) expressed in %. A modest positive association appears between the additional suppression of the lateral channel and top transmission (Spearman $\rho = 0.339$).

Across a large proportion of the parameter space, 2R provides a substantial further reduction in Γside, with many geometries suppressing the remaining one-ring lateral transmission by approximately 30–50% and the strongest reduction reaching approximately 59%. Several negative regions (regions of increased side emission) are nevertheless present, demonstrating that the addition of 2R is not universally beneficial and remains

strongly dependent on the corresponding ring width and separation. A modest positive correlation is observed between increased additional side suppression and Γtop, with a Spearman ρ of 0.339. In clearer terms, geometries in which the introduction of 2R provides greater suppression of the lateral radiation channel appear to preferentially direct a larger proportion of that radiation towards the top channel, and result in an increased ξ as presented in Figure 7.7.

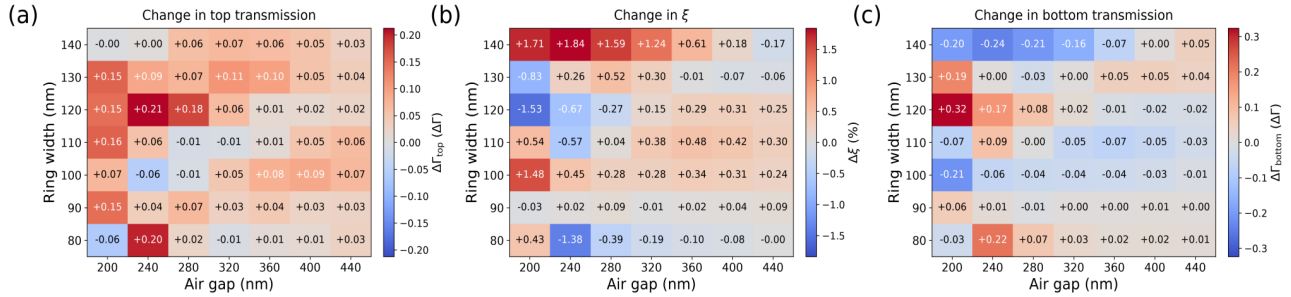


Figure 7.7. Plot of change in (a) Γ_{top}, (b) ξ, (c) Γ_{bottom} between the 1R and 2R geometries. A region of increased ξ is observed, corresponding with a decrease in Γ_{bottom}

7.5.2 Off-Resonant Purcell, ZPL Weight & ξ FWHM

Alongside modifying the emission properties on-resonance (λ_0) , the concentric rings also modify the off-resonant emission. In analysing the off-resonant Purcell factor ($F_{p,+4\text{nm}}$), a detuning of +4nm was selected as the canonical off-resonance point, ($\lambda_{0+4\text{nm}} = 1296.4\text{nm}$). This specific region was chosen as it was sufficiently distant from the cavity response, but still within the region of the QD phonon sideband. As shown in Figure 7.8, the initial $F_{P,+4\text{nm}}$ of 0.55 achieved by the bare micropillar is substantially modified by the addition of 1R & 2R, with regions of enhanced and suppressed $F_{\text{P}+4\text{nm}}$. In the 1R case, geometry clusters cause $F_{\text{P}+4\text{nm}}$ to fall to approximately 0.26, 53% below the bare micropillar value and 42% lower than the corresponding map-mean of 0.45. Alongside this, closely neighbouring geometries can significantly enhance the off-resonant response. Importantly, no significant relationship was observed between off-resonant $\Gamma side$ suppression and $F_{\text{P}+4\text{nm}}$,

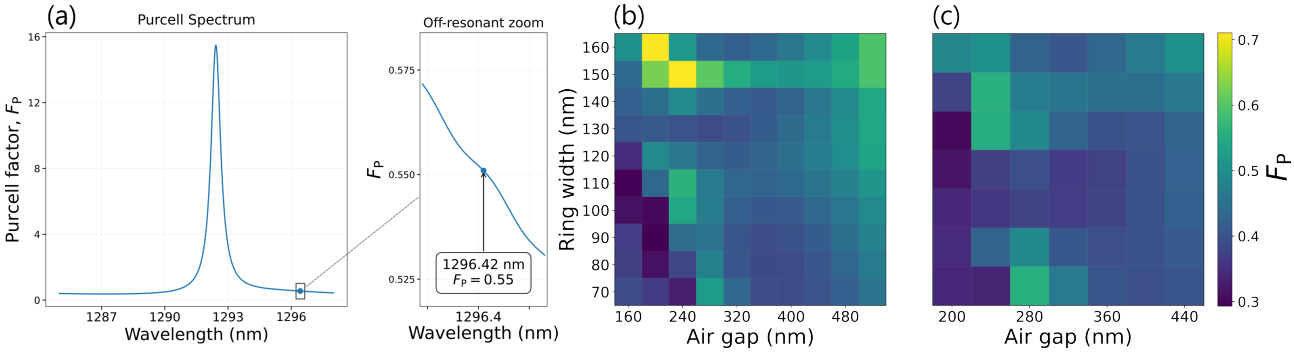


Figure 7.8. Off-resonant Purcell response of the bare and ring-modified micropillars at $\lambda_{\text{res}} + 4\,\text{nm}$. **(a)** Bare-micropillar Purcell spectrum with the off-resonant evaluation point highlighted. **(b)** One-ring 10×10 geometry map and **(c)** two-ring 7×7 geometry map of F_P, plotted using a common colour scale.

meaning that the localised reduction in $F_{\text{P}+4\text{nm}}$ cannot be attributed simply to suppression of the lateral emission channel. A modest positive correlation was, however, instead observed with resonant side suppression ($\rho = 0.382$, $p < 0.001$), suggesting that resonant power redistribution and off-resonant Purcell suppression have distinct responses to the geometric variances.

In summary, despite a strong geometric dependence, the off-resonant response does, in fact, remain below the bare micropillar value across a large proportion of both the 1R and 2R parameter spaces and as such, the addition of concentric rings appears broadly successful in the suppression of unwanted off-resonant radiation. The suppression of the off-resonant F_P is particularly relevant to the PSB, which extends over a significantly broader spectral range than the ZPL Lorentzian. As such, a reduction in $F_{P,+4\text{nm}}$ would be expected to suppress the relative PSB contribution and, consequently, increase the resulting ZPL weight (W_{top}). This increase

in W_{top} is observed, however W_{top} is already close to unity throughout the extent of the investigated parameter space, and therefore only varies minutely over the 1R & 2R geometric configurations. As such, $F_{\mathrm{P+4nm}}$ suppression provides only limited W_{top} improvement to the already highly ZPL-selective top emission channel.

However, a more significant benefit of $F_{\mathrm{P+4nm}}$ suppression is its change to the top extraction efficiency's spectral bandwidth (ξ_{FWHM}). Dlaka *et al.* demonstrated that off-resonant Purcell suppression, combined with a relatively low Q-factor, can lead to a broadening of ξ_{FWHM} beyond that of the cavity resonance itself [6]. The ξ_{FWHM} response was therefore plotted over the 1R & 2R parameter space to compare its geometric response to that of the bare micropillar. As shown in Figure 7.9(a), the bare micropillar has a ξ_{FWHM} of approximately 3.56 nm. Across a large proportion of the 1R parameter space, only a small change is observed in this bandwidth, with ξ_{FWHM} remaining broadly within the range of 3.4–4.0 nm. However, a distinct high-bandwidth region is observed at comparatively small pillar-to-ring separations and intermediate ring widths, as highlighted in Figure 6.9(b). Within this region, ξ_{FWHM} increases to a maximum of approximately 5.87 nm, corresponding to an increase of approximately 65% relative to the bare micropillar. The ξ_{FWHM} response is therefore, like much of the other FoMs, highly geometrically dependent. Whilst a large proportion of the 1R parameter space provides only limited ξ_{FWHM} over the bare micropillar, the existence of a high-bandwidth region demonstrates that the geometry can be tuned specifically towards broadband operation, though this region does not necessarily coincide with the geometries giving the strongest overall combination of resonant Purcell enhancement, lateral suppression and peak extraction efficiency, presenting an additional trade-off.

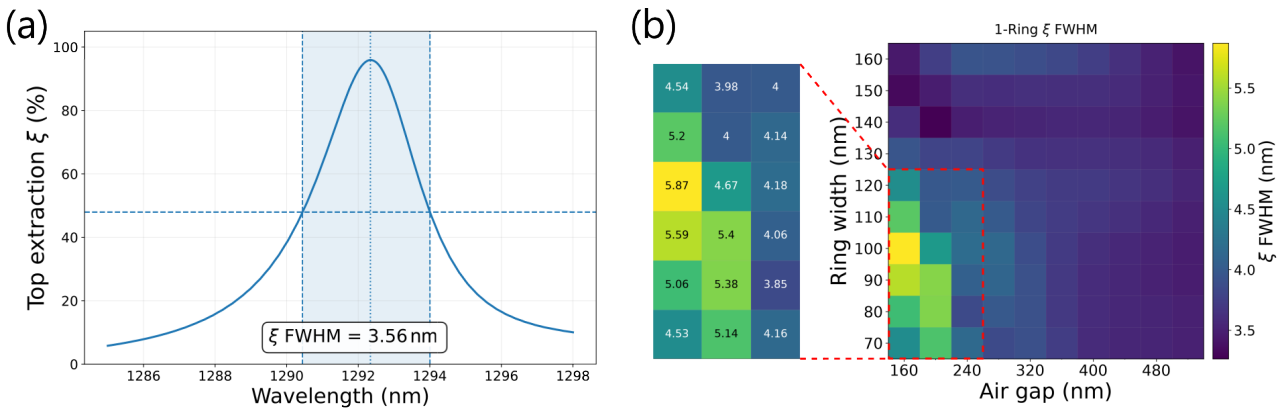


Figure 7.9. (a) Bare micropillar $\xi(\lambda)$ FWHM. $\xi_{\mathrm{FWHM}} = 3.56$ nm. (b) ξ-FWHM across the 1R parameter space. The enlarged inset highlights the high-bandwidth region of the parameter space, where values of up to approximately 5.9 nm are obtained.

7.6 Design Trade-offs

Although much of the investigated 1R & 2R geometries only modestly vary from the bare micropillar, the preceding sweeps do identify several localised geometric arrangements in which individual FoMs are significantly enhanced. These geometry-specific optimums do not tend to coincide with each other, demonstrating a broad, multi-parameter optimisation space in which different geometry configurations can be chosen to selectively maximise different FoMs. To represent these trade-offs, a set of geometries were selected for direct comparison. These were the bare micropillar reference, a balanced BRM geometry (2R, $W_{\mathrm{ring}} = 130$ nm, $G_{\mathrm{air}} = 440$ nm), a broadband/high-extraction geometry (1R, $W_{\mathrm{ring}} = 100$ nm, $G_{\mathrm{air}} = 160$ nm), a maximum side-suppression geometry (2R, $W_{\mathrm{ring}} = 130$ nm, $G_{\mathrm{air}} = 400$ nm), and a geometry for maximum-Purcell enhancement (2R, $W_{\mathrm{ring}} = 130$ nm, $G_{\mathrm{air}} = 400$ nm).

Of all the chosen representative geometries, the broadband/high-extraction geometry presents arguably the most significant divergence from the bare micropillar. ξ_{FWHM} increases from $\xi_{\mathrm{FWHM}} = 3.56$ nm for the bare micropillar case to 5.87 nm, while ξ increases to 96.5%. This increase can be predominantly associated with an observed reduction in $\Gamma bottom$ and consequently improved η rather than stronger lateral confinement. A

Geometry	Rings	W_{ring} (nm)	G_{air} (nm)	F_P	β (%)	η (%)	ξ (%)	ξ_{FWHM} (nm)	$\Gamma side$	$\Gamma bottom$	$F_{P,+4nm}$
Bare micropillar	N/A	N/A	N/A	15.5	98.9	96.7	95.7	3.56	0.15	0.5	0.55
All-rounder	2R	130	440	15.8	99.7	95.1	94.8	3.50	0.05	0.77	0.46
Broadband & high extraction	1R	100	160	15.2	99.3	97.2	96.5	**5.87**	0.1	0.41	0.28
Max Γside **suppression & max** F_P	2R	130	400	**15.8**	**99.7**	94.9	94.6	3.55	0.05	0.8	0.44

Table 7.1. Comparison of representative geometric regimes identified across the BRM parameter space. Each geometry is selected to prioritise a different device characteristic, illustrating the trade-offs between resonant enhancement, directional confinement and spectral bandwidth.

notable drawback of this geometry is that it exhibits comparatively high Γside, alongside a small reduction in resonant Purcell enhancement from 15.8 to 15.1. This presents a trade-off of substantially greater spectral bandwidth and high useful extraction at the cost of increased lateral leakage and slightly reduced spontaneous emission rate.

The maximum Γ*side* suppression geometry instead prioritises lateral leakage suppression. For the 2R configuration at $W_{\text{ring}} = 130\,\text{nm}$ and $G_{\text{air}} = 400\,\text{nm}$, the additional ring produces the strongest reduction in resonant lateral transmission across the parameter space. This regime is desirable where reduction of unwanted lateral radiation is of upmost importance, however, Γside suppression does not directly translate to an equivalent increase in ξ, since the additionally suppressed power is also largely directed towards the bottom channel. Furthermore, this geometry does not exhibit the substantial increase in ξ_{FWHM} observed within the broadband regime (instead, a modest increase is observed, $\xi_{\text{FWHM}} = 3.64\text{nm}$ providing a further trade-off between directional confinement and spectral bandwidth.

7.7 Conclusions and Limitations

The addition of 1R & 2R modifies both the resonant and off-resonant behaviour of the central micropillar while largely retaining its strong F_P enhancement and high ξ. The changes in key FoMs are relatively modest throughout most of the investigated parameter space, however, distinct regions provide substantial improvements in specific FoMs, including geometries with high levels of Γ*side* suppression and others with greatly increased ξ_{FWHM}. There is no single geometry that simultaneously maximises all FoMs, and as such, a clear design trade-off is presented where the optimal geometric configuration depends largely on specific prioritised metrics & use-case. Therefore, the BRM should primarily be regarded as a tunable extension of the micropillar that allows for granular control over the spectral and directional characteristics of the EM field instead of an unequivocal improvement.

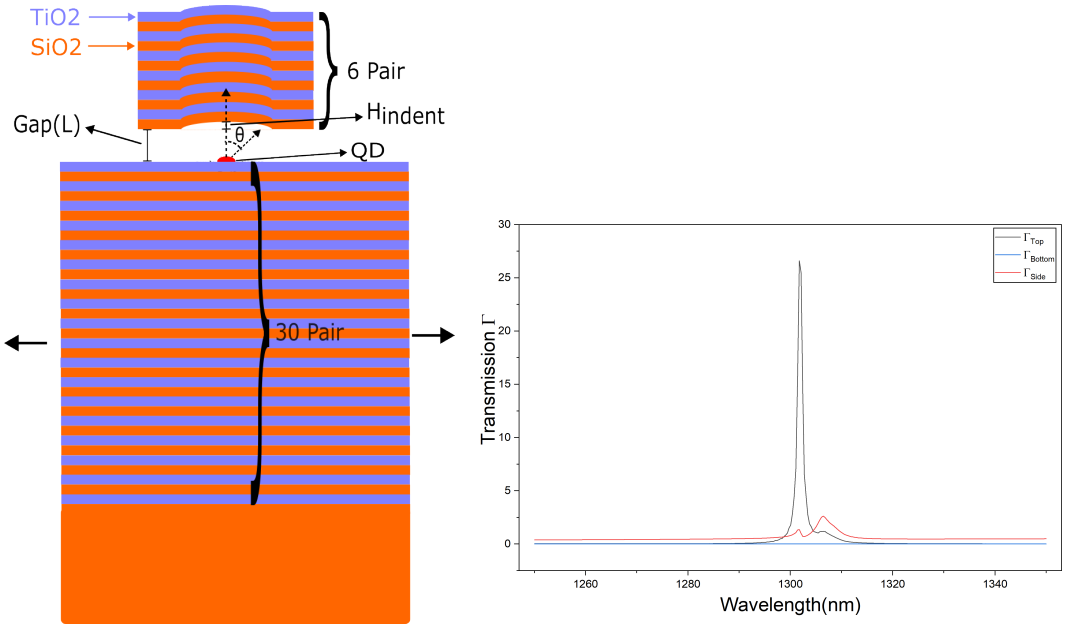
It is highly important to consider the above discussion in the context of several limiting factors. For one, the required trade-off between accuracy & computational resource-requirements of the conducted FDTD simulations meant that a suitable Yee Cell grid density 3.1.1 was chosen which balanced these two factors. As a result, the values extracted from the presented data are prone to a small margin of error and therefore cannot be considered 'absolute'. However, the conclusions in this chapter are drawn predominantly from the trends across the maps (ie. gradual variation in FoM value; for example, the gradual decrease in Γ_{side} over the parameter space). As such, while individual values of quantities like ξ cannot be regarded as fully converged & accurate values, the trends identified across the geometry maps still provide useful insight in spite of this inherent uncertainty.

For the purposes of the comparison (Chapter 9 & 10), the 'all-rounder' (2R $W_{ring} = 130, G_{air} = 440$) architecture was 'carried over' as a representative geometry for comparison with the other structures in this report.

8. Floating open Cavity Architecture(Foc) -JP

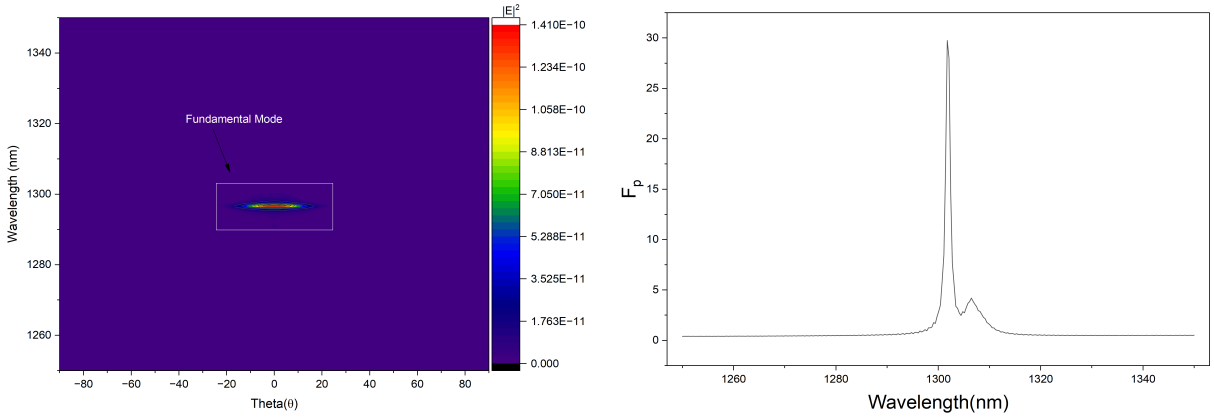
8.1 Architecture and Operating Principle

The Floating open-cavity architecture acts as a Fabry-Pérot resonator formed by two spatially separated reflectors [38] (Fig 8.1) formed by DBR. Unlike micropillars the top DBR stack is a concave cavity suspended in air above the QD. The spectral properties of the light confined and emitted from the structure is affected by a few main parameters: The Gap(L) which represents the distance between the top concave DBR (cDBR) and the bottom DBR, Radius(R) the radius of the concave DBR, H_{indent} the distance from the opening of the cDBR to the top of cDBR. This defines the curvature of the DBR and where the focal point of the cDBR sits and the number of pairs of DBR on the top and bottom. The curvature of the top mirror provides transverse confinement directing the oscillating optical field towards the top of the structure.



(a) Cross-section of Floating open Cavity. Where θ is the lateral confinement angle

(b) The Top,Side and Bottom Transmission(Γ) at H_{indent} = 1.1μm and L = 1μm



(c) Farfield Graph showing the fundamental mode at H_{indent} = 1.1μm and L = 1μm

(d) Purcell Factor(F_p) at H_{indent} = 1.1μm and L = 1μm

Figure 8.1. The structure of cavity and Ideal behaviour in each metric(Top emission(Transmission), Fundamental mode, Purcell Factor(F_p))

This specific geometry dictates the longitudinal resonance via the cavity gap and enforces transverse confinement establishing a localized fundamental Gaussian mode (TEM_{00}). Optimal operation relies on positioning a QD exactly at the longitudinal antinode and transverse mode waist on the planar surface.

The operational advantage of the open-cavity architecture is its geometric tuneability, because the top cDBR is physically separate from the bottom its spatial position can be tuned which allows better top emission and narrow ZPL with low Phonon losses. Further optimising the H_{indent} maximizes coupling efficiency, extraction efficiency and indistinguishability of the generated photons.

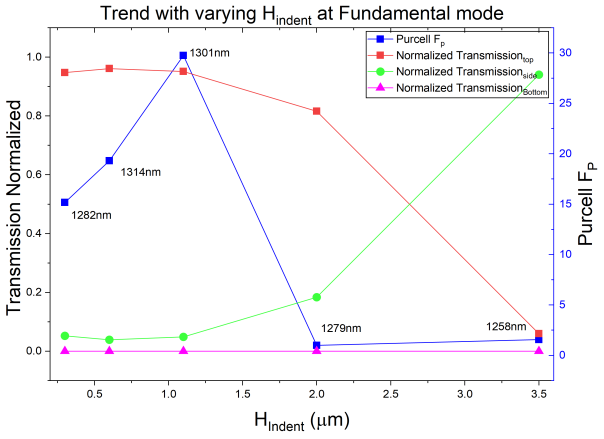
Unlike ideal reflective mirrors, light penetrates several layer pairs into the DBR stacks before being fully reflected. This means the physical gap (L) is shorter than the effective optical length of the cavity. The curvature of the top cDBR must therefore be carefully matched not just to the physical vacuum gap, but to the total effective mode volume that bleeds into both the bottom DBR and the floating top cDBR. Controlling this effective length is what ultimately dictates the longitudinal resonance frequency.

When the quantum dot is perfectly centred at both the longitudinal antinode and the waist of the TEM_{00} mode, it creates an optical environment heavily biased toward emission into the top channel. This spatial overlap between the emitter and the confined electric field is what physically enables the drastic Purcell enhancement explored in subsequent sections, transforming the dot from an isotropic emitter into a highly directional photon source.

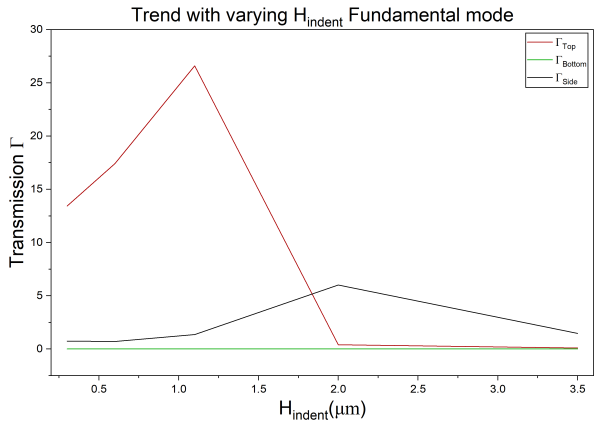
8.2 Sweep and motivation

The primary motivation for mapping out this vast dataset was to understand exactly how the physical depth ((H_{indent}) of the concave mirror influenced both the emission rate of the QD and the escape path of the photons at O-band. Extensive parametric sweeps were executed, systematically stepping through various top DBR indentation depths while tracking the cavity's optical response. Within this sweeping landscape, the initial objective was to hunt down and isolate the fundamental Gaussian mode, filtering out messy higher-order modes that scattered light erratically and suffered from poor transverse confinement.

From an electrodynamic perspective, the emergence of lateral scattering at deeper indentations is governed by diffractive losses. As the cavity indentation becomes too aggressive relative to the gap, the transverse beam waist of the confined Gaussian mode expands. If the size of the optical field begins to exceed the physical boundaries of the machined concave depression, the beam "clips" the edges of the structure. This geometrical mismatch forcefully scatters photons out of the fundamental TEM_{00} mode into leaky, higher-order transverse modes, severely degrading the directional top transmission.



(a) The trend of Purcell and Transmission normalized by $\Gamma_{Channel}/\Gamma_{Total}$ at fundamental mode operation devices across the different depths(H_{indent})



(b) Transmission (Γ) Trend from Transmission monitors setup in Lumerical FDTD at fundamental mode operation devices across the different depths(H_{indent})

Figure 8.2. The above graphs show the trend of Purcell Factor and Transmission Γ as the H_{indent} increases. To ensure fundamental mode operation a Gap(L) sweep was performed at each depth(H_{indent} to avoid higher order modes

However, as the raw transmission trends across these depths were charted, a critical physical trade-off was

uncovered. It was initially assumed that maximizing the total F_p would automatically yield the best device. But when the raw FDTD power was broken down into specific directional channels, the story changed. It was discovered that while a deeper mirror indentation successfully tightened the mode volume and cranked up the overall Purcell, it didn't guarantee more useful light. If the indentation became too extreme, the mirror's focal properties shifted, causing photons to aggressively scatter sideways causing higher order modes that show a high F_p but don't provide useable single photons as there are higher order modes which makes it difficult for light to couple out of the cavity. Therefore analysis were run on sweeps of H_{indent} and Gap(L) to find a sweet spot at each depth. The best device would be the device with Fundamental mode, high top emission and high Purcell Factor(F_p) for ZPL enhancement. By applying these criteria H_{indent} = 1.1μm at Gap(L) = 1μm was selected as the best device from all variations. Shallower cavities than 1.1μm failed to fully maximise the emission rate while deeper cavities have lateral scattering (8.2a).

8.3 Indistinguishability and ZPL

Once the fundamental mode is isolated and locked, the actual quantum quality of the generated photons had to be verified. For applications like quantum computing, mere brightness is not enough the photons must be perfectly identical and indistinguishable. This meant the phonon losses had to be suppressed or the QD would bleed a significant portion of its emission into PSB rather than the useful ZPL.

A Lorentzian was fitted to the F_p to remove the abnormality visible in the Fig:**??**, this ensures there is no unexpected PBS enhancement.To see if the cavity could overcome this, the device operation's were calculated at cryogenic temperature of 4.0 K using the previous Lumerical sweep results. In a standard bulk material, the intrinsic emission is split, with roughly 90% of the light entering the zero-phonon line (ZPL) and 10% bleeding into the broad, decoherent PSB. However, once the cavity perfectly aligned to the 1301.75nm peak, the local density of states fundamentally shifted. The massive Purcell enhancement acted like an optical funnel. For the photons successfully collected by the top mirror, the cavity compressed the emission so rapidly that the ZPL fraction surged to 99.5%. This structure successfully extracted light efficiently and also practically guaranteed pure, indistinguishable single photons.

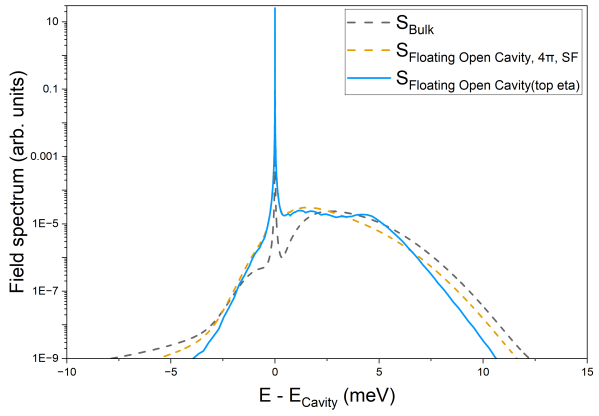


Figure 8.3. Weighted ZPL and PSB enhanced by the Purcell effect at the 1301.8 nm resonance.

Metric	Symbol	Value
Resonance	λ_{res}	1301.8 nm
Purcell Factor	F_p	≈ 29
ZPL Fraction	ZPL	$> 99\%$
PSB Fraction	PSB	$< 1\%$
3LS Purity	$1 - g^{(2)}(0)$	$> 99\%$

Table 8.1. Quantum emission characteristics for the optimized Floating open Cavity (H_{indent} = 1.1 μm, L = 1.0 μm).

By aligning the narrow cavity's resonance with the quantum dot's zero-phonon line, the local density of optical states at 1301.8 nm is drastically amplified. This shifts the system into the Purcell enhanced regime, forcing the radiative decay rate of the exciton to severely outpace the timescale of acoustic phonon interactions. Consequently, the quantum dot emits its photon before thermal lattice vibrations can induce decoherence, physically "funneling" the emission away from the broad phonon sideband and entirely into the transform-limited zero-phonon line.

To formally quantify the single-photon purity the second-order intensity correlation function at zero time

delay, $g^{(2)}(0)$, was calculated. This metric defines the probability of multiphoton emission during a single excitation cycle, where an ideal, deterministic single-photon source yields $g^{(2)}(0) = 0$. The emission purity was evaluated using both two-level (2LS) and three-level (3LS) system dynamics. Because the optimized 1.1 μm cavity achieves a Purcell enhancement of ($F_p \approx 29.2$), the exciton lifetime is compressed from a bulk value of 1000 ps down to approximately 34 ps. This rapid emission virtually eliminates the probability of the dot being pumped twice by the same laser pulse. Consequently, the 3LS estimation yields a $g^{(2)}(0)_{3LS} < 0.01$, corresponding to a single-photon purity exceeding 99%.

To further evaluate the structure's performance figures of merit [Ch:3] were extracted. For the selected best variation of the Foc which is H_{indent} of 1.1μm and L of 1μm at the resonance wavelength 1301.75nm. The β reaches 95%, indicating the vast majority of the QD's emission successfully couples into localized cavity mode rather than leaking laterally into the environment. Further the η exceeds 99%, the bottom DBR functions as a perfect reflector directing all coupled light upward. This results in ξ of 95%, which shows almost all of the photons escape from the top.

8.4 Conclusions and Limitations

Through systematic FDTD simulations, the theoretical advantages of the Floating open cavity architecture were demonstrated. By optimizing the Gap(L) and cDBR depth (H_{indent}) to a geometric sweet spot of an indentation depth of 1.1 µm and a gap of 1.0 µm, the structure successfully channeled a massive 29-fold emission enhancement cleanly into the fundamental mode at 1301.8 nm. This optimal configuration yielded β of roughly 95% and a near-perfect η of over 99%, culminating in an overall ξ of approximately 95%. Furthermore the rapid 34 ps emission effectively suppressed the phonon sideband, guaranteeing a single-photon purity exceeding 99.7%. These metrics confirm that, strictly from an optical standpoint, the floating open-cavity is an exceptional architecture for deterministic quantum light generation.

However, the very feature that enables this remarkable optical performance—the physical separation of the bottom DBR and top cDBR is simultaneously its greatest engineering limitation. The open gap makes the cavity acutely sensitive to mechanical vibrations and thermal drift. The vulnerability of the floating cavity ultimately stems from the fundamental physics of optomechanical coupling. In a high-finesse Fabry-Pérot resonator, the optical resonance frequency is dependent on the exact macroscopic length of the gap (L). Consequently, even sub-nanometer mechanical vibrations—driven by acoustic noise or cryogenic cold heads translate directly into severe spectral jitter. This rapid microscopic shifting of the cavity resonance broadens the effective optical linewidth, periodically detuning the cavity from the quantum emitter and drastically degrading the average Purcell enhancement over time, making active stabilization a strict physical necessity.

To actually realize these simulated extraction efficiencies and exceptional zero-phonon line purities in a physical laboratory, the top cDBR cannot simply hover it must be actively stabilized. This requires integration with complex, cryogenically compatible piezoelectric positioners capable of nanometer-scale precision[39]. Ultimately, the parameter sweeps proved that while the optical performance and spectral tunability of a floating open cavity can far surpass static monolithic devices, it demands an extraordinary level of dynamic mechanical control. The quantum dot must remain perfectly locked at the optical antinode to prevent lateral scattering and maintain the spatial confinement required for highly pure single-photon emission.

9. Quantitative Comparison - CB, JP, MM

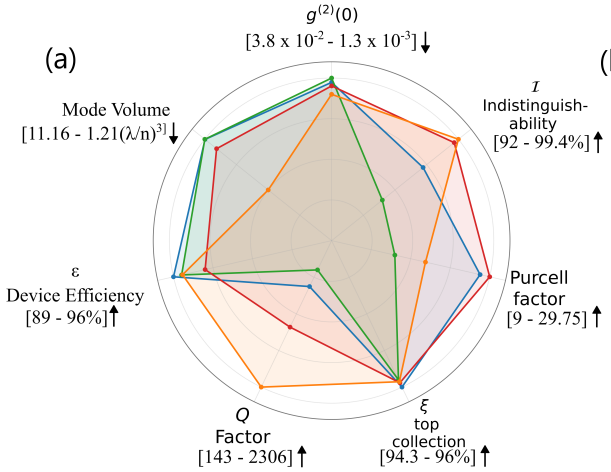
This chapter presents the selected structures studied throughout this report compared on key identified FoMs. The comparison includes the two CBG structures, the FoC, and the BRM. The objective for this section is to establish the performance trade-offs between the structures, before clarifying the best applications in 10.

9.1 Selected FoMs

A common set of figures of merit (FoMs) is used to provide a quantitative comparison between the device architectures. These FoMs comprise of the Purcell factor F_P, cavity quality factor Q, normalised mode volume $V/(\lambda/n)^3$, top collection efficiency ξ, device efficiency (ϵ), the second-order purity $g^{(2)}(0)$, and indistinguishability (I). These figures were selected specifically to model the trade-offs between the different structures and visualise the strengths and weaknesses of each.

9.2 Spider Diagram

Figure 9.1 summarises the relative performance of the selected structures across the selects FoMs defined above. Panel (a) presents the normalised spider diagram, while panel (b) gives the corresponding physical values used to construct the comparison. Each spoke is normalised independently so that the differences between structures remain visible even for metrics that occupy very different numerical ranges. As such, the radial distance on one spoke should not be compared directly with the numerical scale of another. For all axes, being further out on the spoke represents more a higher performance. The $g^{(2)}(0)$ and mode-volume (V) axes are therefore reversed, since smaller values are generally preferable for both quantities.



(b)

	CBG Blue Sky	CBG Green Grass	FoC	BRM 2 Ring
$g^2(0)$	1.1×10^{-2}	1.3×10^{-3}	1.9×10^{-2}	3.8×10^{-2}
Indistinguishability I (%)	96.0%	92.2%	99.0%	99.4%
Purcell Factor F_p	27.65	9	29.75	15.7
Internal Efficiency ξ (%)	96.0%	94.3%	95.0%	94.8%
Device Efficiency ε (%)	**96.0%**	**94.2%**	**89.0%**	**94.0%**
Q-factor	450	143	1199	2306
Mode Volume $(\lambda/n)^3$	1.24	1.21	3.06	11.16

Figure 9.1. Comparison of the structures using selected FoMs. (a) Normalised spider diagram comparing $g^{(2)}(0)$, $\mathcal{I}$, F_P, ξ, Q-factor, ϵ, and normalised mode volume. All axes are oriented such that greater radial extent represents better performance. The $g^{(2)}(0)$ and mode-volume axes are reversed, with lower physical values plotted further from the centre. (b) Corresponding physical values used to construct the spider diagram for each structure. An up arrow (↑) denotes values increasing in radial distance. A down arrow (↓) denotes the opposite, as seen in the cases of mode volume & purity.

9.2.1 Mesh-Order Verification

Representative final geometries were re-simulated at mesh-order 5 (M5) to verify the mesh order 3 (M3) simulations used throughout this work. Across the open cavity, CBG and BRM structures, the performance metrics changed only modestly. For example, F_P changed from 29.75 to 29.23 for the open cavity, from 30.87 to 32.80 for Blue Sky, and from 15.7 to 15.99 for the BRM, while directional efficiencies remained similarly stable. The larger mesh dependence was generally observed in the precise resonance wavelength rather than in F_P, Q or extraction behaviour, supporting the use of M3 as a computationally efficient resolution for the parameter sweeps and comparative conclusions described in Chapters 3–8.

10. Applications MPW & CB

Figure 1.1 in Chapter 1 introduced the single-photon source as the photon-generation stage within a wider quantum-photonic system, with quantum computing, quantum communications and quantum sensing shown as three representative application areas. The preceding chapters established how the investigated architectures operate, and Chapter 9 compared their performance on a common basis. Subsequently, this chapter compares the representative structures using the FoMs defined in Chapter 9. The W1 photonic-crystal waveguide is excluded from this comparison due to the additional challenges faced during simulations, discussed in further length in Chapter 8 Section 4.5.

10.1 Fabricability & Engineering Trade-offs

The conducted FDTD simulations, while accurate for electromagnetic modelling, fail to account for real-world fabrication imperfections such as QD-misalignment, etch-imperfections, and variability in grown wafer dimensions (covered in greater detail in Chapter 6 Section 3.1.4) and instead provide something of a 'best case' simulation, where all physical parameters are at their optimum (ie. perfect QD positioning, no sidewall roughness, no uncertainty in geometric proportions). As such, fabricability remains a key question when selecting a suitable device/devices for each potential application and must be taken in to consideration:

Architecture	Fabrication method	Advantages	Drawbacks
Micropillar / BRM	Epitaxially grown Micropillar with subsequent etching for Bragg-rings	Compact monolithic architecture allows for easier fabricability	Rings introduce additional geometry prone to fabrication imperfections
CBG	Concentric rings of a chosen medium atop a semiconductor, separated by an etched air gap	Deterministic QD-cavity alignment and efficient free-space extraction have been demonstrated	Device performance depends on the uniformity of all ring-widths and air-gaps. The more rings there are, the more potential fabrication variations.
FoC	QD-containing semiconductor coupled to an adjustable opposing mirror	Positioning of the external mirror can compensate for imperfect QD location	Additional alignment required, small variations in concave mirror induce losses.

Table 10.1. Fabrication characteristics of the selected structures included in the quantitative comparison

For the BRM architectures, deterministic fabrication around pre-selected QDs is experimentally established [37]. Where the authors, Madigawa et al., measured a collection improvement substantially smaller than the idealised prediction and attributed the shortfall to fabrication deviations [37]. The CBG, however, has strong published evidence for scalable deterministic fabrication. Barua et al. fabricated 103 QD–CBG devices across two sessions, of which 95 were optically active, giving a yield $> 90\%$ [40]. Those devices operated near 895 nm on a different stack and geometry, so the figure demonstrates that high-yield fabrication is attainable, but it is important to note that this figure cannot be read as a predicted yield percentage for 1300nm structures investigated in this project. Additionally, Chapter 6 adds that DBR-assisted variants (both suspended in-air and non-suspended) impart additional fabrication complexities, particularly in the case of CBG Blue Sky, which is presented as an idealistic case. The FoC trades fabrication-fixed resonance for post-fabrication adjustability. Hoang et al. reported no measurable vibration induced broadening for a cavity with a finnesse above 1000 operating in a standard closed-cycle cryostat without active locking, with approximately 3 nm of tuning at a resonance λ of 971 nm [41]. While this is not directly applicable to the 1300 nm structures, what is demonstrated is the feasibility of post-fabrication tuning of the suspended micromirror. The resulting burden is an intensive process of assembly and alignment of the aforementioned micromirror, alongside the drawback that multiple FoCs require multiple of these positioning assemblies, compared to – for example – a micropillar array in which several structures can be placed close together, without the need for careful post-fabrication alignment and maintenance.

10.2 Photonic Quantum Computing (PQC)

Photonic Quantum Computing (PQC) is an optical regime of Quantum Comptuing (QC) in which photons are used as qubits, the quantum equivalent of classical computational bits (0, 1) [42]. Qubits leverage the multiple degrees of freedom of a photon (including polarisation, spatial, temporal and frequency) to encode quantum information. Chapters 1 and 2 established that PQC requires a combination of high device efficiency, high purity and high indistinguishability at the same time. Ding et al. demonstrate that these can be achieved jointly in a resonantly driven QD micropillar source [43]. Given that suitability is dependent on all aforementioned FoMs (ϵ, $g^{(2)}(0)$, I) being highly optimised, a collated FoM was devised to assess each device's overall performance for the application of PQC, defined as ζ_{PQC} in Equation 10.1 below:

$$\zeta_{PQC} = \epsilon\, I\, \mathscr{P}, \qquad \mathscr{P} = 1 - g^{(2)}(0) \tag{10.1}$$

Where ϵ is device efficiency (Chapter 3 Equation 3.3.4), I denotes Indistinguishability (Chapter 3 Equation 3.3.2), and $\mathscr{P}$ denotes single-photon purity (Chapter 3 Equation 3.14. As such, the probability of attaining a certain number of correct photonic qubits from the emitter can be modelled as:

$$S_{\mathrm{PQC}}(N) = \zeta_{\mathrm{PQC}}^{N} = [\epsilon\, I\, \mathscr{P}]^{N} \tag{10.2}$$

Where N denotes the number of qubits (assuming one photon per qubit).

(a)

Structure	ζ_{PQC}
CBG Blue Sky	91.2%
BRM 2R	89.9%
CBG Green Grass	86.7%
FoC	86.4%

(b)

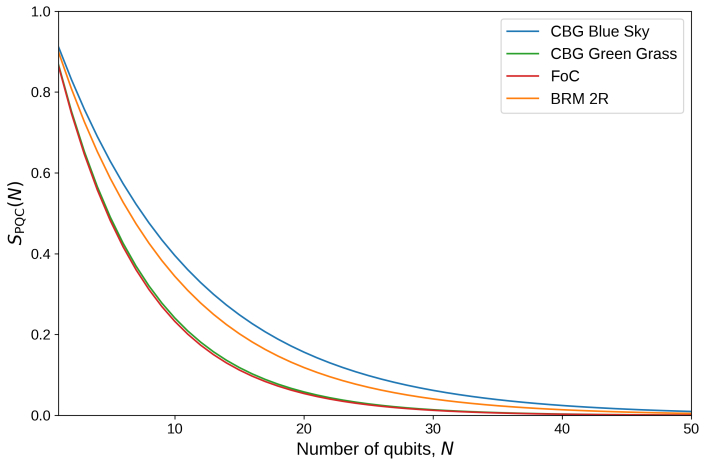


Figure 10.1. (a) Composite PQC figure of merit, ζ_{PQC}, for each structure (b) Scaling of $S_{\mathrm{PQC}}(N)$ with qubit number.

There are two clear highest performers, the BRM 2R $\zeta_{PQC} = 89.9\%$ and CBG Blue Sky $\zeta_{PQC} = 91.2\%$. Given that the Blue Sky is presented explicitly as an idealistic case, its use must be considered with future fabrication challenges (outlined further in Chapter 6) in mind. As such, for present-day, practical fabricability, the BRM 2R is the clear winner, with mostly comparable (though slightly diminished) performance in qubit-scaling but with a significantly clearer route to fabrication. If disregarding the Blue Sky's unresolved fabrication limitations and assuming assembly can be achieved without significantly affecting the fabricated device's performance, its greater $S_{PQC}(N)$ performance as N increases makes it a more suitable candidate, purely on a performance basis.

10.3 Quantum Key Distribution (QKD)

Quantum Key Distribution (QKD) sits within the paradigm of Quantum Networking as a method of establishing a shared encryption key between two parties while allowing eavesdropping attempts to be reliably detected [44]. The term QKD itself is something of an umbrella, with multiple sub-protocols placing different demands on the SPS FoMs. In this section, two relevant QKD classes are considered. The first is prepare-and-measure QKD (PM-QKD), where photons are prepared in selected quantum states before being sent to a receiver for measurement. As such, the photon acts as a carrier for encoded key information (a notable

example being the BB84 protocol [45]). The second is measurement-device-independent QKD (MDI-QKD), an interference-based system in which two parties send photons to a central measurement node [46]. For successful two-photon interference, these photons must be closely matched spatially, temporally, and in frequency [47]. As such, PM-QKD and MDI-QKD require strong performance in different FoMs, which are outlined in Table 10.2 below:

QKD Class	Most Important FoMs	Requirements
Prepare-and-Measure (PM-QKD)	Low $g^{(2)}(0)$	High single-photon purity and efficient generation/extraction of photons for key transmission.
Measurement-Device Independent (MDI-QKD)	High I, low $g^{(2)}(0)$	Similar to the requirements of PM-QKD, but additionally requiring indistinguishability for two-photon interference.

Table 10.2. Key single-photon source figures of merit for prepare-and-measure QKD (PM-QKD) and measurement-device-independent QKD (MDI-QKD). Alongside the listed important FoMs, high ε & high F_{P} also hold weight as more general desirable characteristics for SPSes.

For PM-QKD, CBG Green Grass and CBG Blue Sky represent the strongest candidates. Green Grass achieves the lowest $g^{(2)}(0)$ of 1.3×10^{-3}, almost an order of magnitude stronger that Blue-Sky ($g^{(2)}(0)$ of 1.1×10^{-2}), and is therefore the strongest structure when maximising solely for the highest purity. This does, however come at a cost of a substantially reduced F_P of 9, compared with 27.65 for Blue Sky (a F_{P} corresponding with a higher rate of spontaneous emission) . Blue Sky additionally achieves the highest device efficiency and top collection of 96%, making it the stronger compromise where both photon purity and emission rate are considered. Again, fabricability concerns remain a distinct factor 6.5. Green-Grass may therefore be regarded as the stronger purity-specialised source, with a lower overall emission rate (and thus bit-rate) while Blue Sky provides higher spontaneous emission rate enhancement and comparable useful output at the cost of additional fabrication challenges.

For MDI-QKD, the additional requirement of high indistinguishability makes both CBG architectures less suitable and naturally favours both the BRM and FoC. Both structures report similar, highly maximised indistinguishabilities ($I_{BRM} = 99.4\%$, $I_{FoC} = 99\%$), while the FoC has a purity approximately twice as strong as the BRM ($g^{(2)}(0)_{BRM} = 3.8 \times 10^{-2}$, $g^{(2)}(0)_{FoC} = 1.9 \times 10^{-2}$), as well as a significantly higher rate of spontaneous emission ($F_{P_{BRM}} \approx 15.7$, $F_{P_{FoC}} \approx 29$. Despite this, the real trade-off between these structures is seen in implementation practicality. BRMs are significantly easier not only to fabricate, but also to mass-produce (with multiple high-performing devices able to be placed in close proximity on the same substrate) [37], whereas FoCs require significant post-fabrication alignment & tuning; a significantly higher engineering burden.

As such, while a high-performing FoC outperforms a BRM on several important MDI-QKD metrics, the BRM's higher comparative ease-of-implementation, alongside its higher device efficiency ($\epsilon_{BRM} \approx 94\%$, $\epsilon_{FoC} \approx 89\%$) make it a significantly more achievable solution, with higher potential for mass-production and cost-effective maintenance.

10.4 Quantum Sensing & Metrology (QSM)

One concrete application of quantum metrology is quantum radiometry. Because quantum radiometry deals with optical signals so weak that individual photons are detected one-at-a-time, it serves as an instructive use-case for quantum metrology in general. When presented with a signal this weak, the detector sees photon arrivals instead of a continuous signal. In this situation, a well-calibrated SPS can offer significant benefit: because each trigger attempts to release a photon with some probability P, the photon input presented to the detector can be statistically characterised. Kück et al. cite photon flux, single-photon purity, and spectral power distribution as relevant characteristics of sources in this application, including semiconductor QD SPSs [48].

Photon-number purity is relevant because the calibration assumes a controlled photon input: if the excitation event creates two photons instead of one, the detector sees more light than expected based on the calibration.

If that extra photon triggers a count, the calibration corresponds to a higher-than-intended photon flux. For this reason, low $g^{(2)}(0)$ is required, making Green Grass a strong candidate. Against this metric, Green Grass clearly outperforms the other free-space architectures considered. With a modelled $g^{(2)}(0) = 1.3 \times 10^{-3}$, it is 8.5 to 29× lower than any other free-space SPS. This advantage is achieved while device efficiency remains near 94%. The fabrication assessment made in Section 10.1 favours Green Grass for this reason as well, since Green Grass is one of the more feasible designs.

This recommendation does not necessarily apply to quantum sensing in general, since sensing can use photons in different ways. In interferometric sensing, one measures a quantity by how it changes the interference of optical fields/photon wavefunctions. Hong-Ou-Mandel quantum microscopy implemented by Ndagano et al. [49] is an example. Where the photons interfere, their indistinguishability strongly influences the visibility of that interference. For wavelength-sensitive sensing, the emission wavelength must match the desired transition, which could favour a tunable source like the FoC. Metrology often requires long-term source stability or absolute flux, neither of which was modelled here. These properties would have to be measured in order to complete this assessment.

10.5 Summary

Chapter 1 established that there could not be a single figure of merit to identify a useful SPS. Useful optical modes must be populated with photons, additional photons must be suppressed, and the optical properties that the intended receiver apparatus depends upon must be preserved. Which of those three properties matters most in a given situation depends on the application. Quantum computing requires photons in a different capacity than quantum communications and quantum sensing. Due to those application-dependent priorities, the ranking process of Chapter 9 and fabricability assessment of Section 10.1 combine to prescribe a methodology for selection based on application.

Chapter 9 stressed that several figures of merit must be considered when choosing an SPS. Device efficiency, photon-number suppression, and indistinguishability of separately emitted photons are all important to its function. Blue Sky was highlighted throughout this text to serve as a point of comparison for this conclusion. While it may have strong modelled optical performance in the ideal scenario, the air-gap support mechanism is currently too unresolved for Blue Sky to be reasonably selected. Simulation results cannot be treated as demonstrated device performance while such a significant structural uncertainty remains.CBG Green Grass was determined to be the preferred choice when photon-number control is the application driver. Prepare-and-measure QKD is hurt by additional photons because they alter the single-photon state input into the encoding procedure. The same is true when using photon-number-sensitive metrology to calibrate a detector, when said detector is used with a reference optical source. If an additional photon reaches the detector during calibration, the expected count rate is altered. Since this application requires photon number purity the most, Green Grass was determined to be the best free-space option. It offers the lowest modelled $g^{(2)}(0)$ and does so while keeping high useful output. Finally, a deterministic fabrication route has been demonstrated for similar CBG devices, so selecting Green Grass is encouraged by factors outside of its modelled source metrics.

MDI-QKD once again changes the application requirements. Interference is required between independently generated photons at the central measurement station. In order to interfere, the two optical states must be sufficiently well matched. Spectral alignment and indistinguishability must be taken into account when discussing interference. Due to both its high modelled indistinguishability and post-fabrication spectral tuning, FoC becomes the design of choice for MDI-QKD; the spectral tuning allows for direct compensation of variations between independently grown QDs, with the BRM more suitable on a practicality-basis. The rest of the structres provided similarly useful conclusions. Every device architecture has its own strengths and weaknesses, leaving the decision of what to pick as a matter of which application each architecture is best suited to. For the application of PQC, this work suggests the BRM 2R. For PM-QKD and photon-number-sensitive metrology, the CBG Green Grass was found to be the best fit. For interference-based MDI-QKD, the FoC was determined to have the right combination of qualities to be used as the preferred source.

11. Conclusion - IC & AQ

Single-photon emission alone does not define a useful source. The photon must reach the required optical channel, multiphoton emission must be supressed and successive photons when required must remain optically indistinguishable.Five nanophotonic architectures were therefore evaluated according to their suitability for applications and ease of fabrication rather than any single FoM. No single architecture was found to be universally better than all others.

11.1 Architecture-Specific and Cross-Architecture Findings - AQ, MPW

Narrowing the micropillar resonance with additional upper DBR pairs, and 13TP led on Purcell enhancement, useful upward fraction, ZPL fraction and model-derived indistinguishability. The shorter lifetime also enabled increased re-excitation within the pump pulse, so the 7TP offered the highest photon-number purity.

The CBG acts on where emission goes instead of the rate it leaves: concentric trenches redistribute radially propagating light, and the lower DBRs suppressed radiation directed into the substrate. Downwards leakage was nearly fully suppressed by Green Grass, while simultaneously retaining high upward output, a broad response and the lowest modelled multi-photon contribution of the CBG designs. Full etching and a controlled membrane-DBR air gave enabled stronger confinement in Blue Sky, along with Purcell enhancement and spectral purification, but with a trade-off of greater modelled re-excitation penalty and demands to fabrication which remain unsolved.

Vertical resonant confinement is already provided by the 13TP bare micropillar, so the surrounding rings were introduced principally to act on lateral radiation. Their width and spacing strongly affected the response, with reduced lateral leakage across some of the swept geometries, and one region broadened the useful extraction bandwidth by approximately 65% in connection to the bare micropillar. Because to the additionally suppressed lateral power was not consistently redirected upward, and because regions of enhanced FoMs were observed at varying geometric regimes, the BRM is better viewed as a tuneable extension of the micropillar rather than an unequivocally superior replacement.

Separate mirrors permitted the FoC to be tuned after fabrication, so independent QD emitters can be brought into spectral alignment. The selected design combined directionality and strong simulated enhancement with high model-derived photon quality, but at the cost of greater alignment and mechanical-control demands.

Emission in the W1 PCW is coupled laterally into a guided in-plane mode rather than into a free-space vertical output as in the other four architectures. Converged designs showed strong Purcell enhancement and guided mode coupling, but with the downside of the resonances so narrow that the fields take a long time to decay away, which limited convergence of the longer FDTD simulations, and the resonance remained near 994 nm rather than the targeted O-band of 1260 nm to 1360 nm. With several free-space metrics not directly equivalent for guided output, the waveguide was excluded on the grounds of comparability, not for weaker performance.

Geometry therefore governed not only the magnitude of the enhancement, but consequently the rate of emission, its direction, its useful spectral range and the retained photon quality.

11.2 Application-Aware Engineering Conclusions - AQ

For PM-QKD and photon-number-sensitive metrology, the Green Grass CBG could be chosen. Its low multiphoton probability, high useful output and comparatively credible fabrication route met the demands of high photon-number purity. MDI-QKD instead favoured the FoC, as its high predicted indistinguishability and post-fabrication spectral tunability were particularly useful for matching photons emitted from independent quantum dots. In PQC, the choice must be made between idealised performance and the feasibility of engineering implementation. Blue Sky achieved the stronger idealised combination of optical performance, while

the BRM provided a broadly comparable and experimentally grounded route.

11.3 Limitations and Confidence in the Findings - AQ, MPW

All the FDTD models used in this project were idealised, assuming perfect device geometry, perfect emitter placement, and no fabrication disorder. Meanwhile, Finite mesh resolution and simulation duration also limited the absolute numerical accuracy of the results.. In addition, the architectures did not always possess directly equivalent physical output channels, particularly when comparing the PCW with the vertically emitting cavity structures. Source-level quantities such as $g^{(2)}(0)$ and indistinguishability were model-derived rather than obtained directly from FDTD and therefore depended on assumptions regarding emitter lifetime and excitation, the treatment of phonon effects, and the omission of separately modelled pure dephasing and slow spectral diffusion. Furthermore, no devices were fabricated or experimentally characterised within this work. The final conclusions should therefore be interpreted primarily as comparative design trends and application-aware engineering predictions rather than predictions of absolute experimental device performance.

11.4 Future Work & Considerations - IC

Experimental validation should be a priority to check whether fabricated devices retain the trends observed in the simulations. Then, future experiments could focus on a few representative devices, measuring their spectra, radiative lifetime and collection efficiency. We could also look at photon statistics and two-photon indistinguishability if enough experimental data are available. Future simulations would also include realistic fabrication effects, such as QD displacement, dimensional variation, etch uncertainty and device supports. For collection, finite-NA or fibre-mode coupling could give a more realistic measure than the upward power calculated in FDTD. For the W1 PCW, further work could complete the comparison at O-band. If the long decay times make FDTD impractical, methods like eigenmode or frequency-domain could be used instead. In addition, using experimentally measured dephasing and spectral-diffusion parameters may also improve the source model.

11.5 Final Conclusion

No single architecture could be considered superior within the investigated design space, as each structure was better suited to different quantum applications depending on the required figures of merit and engineering constraints. Throughout the project, Building on the physical principles established in chapter 1 and 2, and using the FDTD and metric-extraction methodology defined in chapter 3, device geometry determined the electromagnetic response, which in turn influenced the emitted-photon properties and their suitability such applications discussed in chapter 10. Consequently, the structure providing the greatest Purcell enhancement was not necessarily the most useful single-photon source. The principal outcome of this project was therefore the optimisation and comparative evaluation of the investigated QD Nanophotonic structures, enabling the most suitable architectures to be identified for different quantum applications according to their required figures of merit and practical engineering constraints.

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